Timing undernutrition: How Survey Timing Distorts Child Growth

Habibou Ibrahim Kassoum*

October 2026

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* Université Clermont Auvergne, CNRS, IRD, CERDI, F-63000 Clermont Ferrand, France

Abstract

Reducing child undernutrition and tracking progress toward that goal rely on anthropometric data collected during household visits. Using quasi-random variation in the within-day timing of interviews across Sub-Saharan African countries, this paper shows that recorded weight among children under five increases over the course of a day. As a result, children measured after 4pm are 19 percent less likely to be classified as wasted, and 16 percent less likely to be classified as underweight than similar children measured in the morning. These findings suggest that about 1.06 million undernourished children across the region are missing from official statistics, and a further 338,000 are recorded as moderately rather than severely undernourished. These results have important implications for research on child undernutrition and for monitoring progress toward its reduction.

1. Introduction

Child undernutrition is one of the major challenges in Low and Middle-Income countries (LMICs) and contributes to nearly half of all deaths among children under five (Black et al., 2013). For children who survive, research shows that nutritional deficits during early childhood translate into fewer years of schooling, poorer cognitive development, shorter adult stature, poorer labor market outcomes and lower adult earnings (Alderman, 2006; Case & Paxson, 2006; Glewwe & Miguel, 2007; Schultz, 2002; Strauss & Thomas, 1998; Victora et al., 2008). The fetal development and early-life health literatures place these effects in a broader frame, showing that nutrition shocks in the first thousand days affect adult health, disability and mortality (Almond & Currie, 2011; Currie & Vogl, 2012; Maccini & Yang, 2009). At the macroeconomic level, undernutrition costs LMICs an estimated 2 to 3 percent of GDP each year, a loss comparable to the annual growth rate of many of these economies (Shekar et al., 2006).

Monitoring progress toward global child nutrition targets relies on anthropometric data collected through large-scale household surveys. Under Sustainable Development Goal 2.2, adopted in 2015, governments committed to ending all forms of child undernutrition by 2030, establishing a measurable policy target and a framework for financing and evaluating nutrition interventions. In practice, progress is tracked primarily through large-scale household surveys, especially the Demographic and Health Surveys (DHS) and the Multiple Indicator Cluster Surveys (MICS), two programs that between them have fielded more than eight hundred surveys across some one hundred and twenty countries since the mid-1980s. In these surveys, trained enumerators measure the weight and height of children under five, and these observations are used to build national estimates of stunting, wasting, and underweight, which feed into the Joint Child Malnutrition Estimates and SDG monitoring systems.

Beyond tracking undernutrition, anthropometric data are also important for identifying which interventions improve child health and nutrition. Studies use these measurements to evaluate the effects of deworming and iron supplementation on child health and growth (Baird et al., 2016; Bobonis et al., 2006; Miguel & Kremer, 2004), school feeding and nutrition information (Afridi, 2010; Fitzsimons et al., 2016; Kazianga et al., 2014), cash transfers and early childhood programs on child development and growth (Attanasio et al., 2020; Gertler, 2004; Macours et al., 2012), and women's empowerment on child nutrition (Duflo, 2003; Lépine & Strobl, 2013; Qian, 2008). The findings of these studies inform the allocation of nutrition budgets and the design of social programs across LMICs. Progress against child undernutrition therefore depends on two kinds of evidence, good survey data to track how many children are undernourished, and good research on child interventions to establish what works. Both rely on the accurate measurement of children’s weight, height and age, making the quality of anthropometric measurements important for research and policy.

The accuracy of anthropometric indicators depends on the quality of children’s recorded weight, height and age, yet each step is subject to error. Children should be weighed in minimal clothing, yet clothing or a wet nappy can add mass to the reading of about 80 grams in hot weather to nearly 800 grams in cold (Tuan et al., 2002). Height is subject to more errors, as length must be measured lying down on a recumbent board until 24 months of age and standing thereafter, yet boards are often not available in the field or avoided by enumerators who find them bulky and fear distressing the child (Mwangome et al., 2012; Mwangome & Berkley, 2014). When a child is measured in the position meant for the wrong age, the two protocols differ by roughly 0.7 centimeters, and because the standard correction is often not applied, the child's height can be misstated (Buyken et al., 2005). Movement and bent knees can affect recumbent measurements, while shoes, hair, and poor posture can affect standing measurements (WHO, 2006). Even trained enumerators differ from one another, with inter-observer error ranging from 0.04 to 2.58 centimeters in a large Ethiopian survey (Moss et al., 2020). Age introduces another

important source of error. In Sub-Saharan African (SSA), only about 44 percent of children under five have a registered birth (UNICEF, 2013), and ages are often estimated using memorable events or rounded to whole years producing the familiar heaping at 12, 24 and 36 months (Chayovan & Knodel, 1993; Fayehun et al., 2020). Month of birth is often misreported, and non-response on child age reaches 30 percent in some DHS (Larsen et al., 2019). Because anthropometric z-scores depend on the relevant measurements, sex, and age, errors in these inputs can affect both the resulting scores and children's nutritional classification (Ulijaszek & Kerr, 1999). [1]

This paper introduces the time of day at which children are measured as a source of a systematic bias. The most direct evidence comes from Wright et al., (2025), who repeatedly collected weight data for children under three years old over three days. Among toddlers (aged 12 to 36 months) weight fell overnight and increases through the day, whereas infants (aged under 12 months) showed no day-night difference. The authors attribute this difference to the feeding patterns as infants take milk in frequent small amounts around the clock, so their weight is roughly unchanged over the day while weaned toddlers eat discrete meals in daylight and fast overnight so gut contents and hydration accumulate from morning to evening increasing their recorded weight. Eating in the previous two hours was also associated with an increase weight. Other studies show that short-term weight fluctuations can be large relative to true growth (Wright et al., 2022), that ingested milk registers directly on the scale (Savenije & Brand, 2006), and that comparable day-to-day fluctuation occurs among Gambian infants (Naumenko et al., 2021). Because enumerators visit households throughout the day, two otherwise identical children may be recorded with different weights solely because of when they were visited. Child are classified as underweight, wasted, or stunted if their weight-for-age (WAZ), weight-for-height (WHZ), or height-for-age (HAZ) falls more than minus two standard deviations below the international reference median (WHO, 2006). Because weight can change over the day, children measured at different times can register different WAZ and WHZ, so some children near the threshold are classified as undernourished or not depending solely on when they were measured, and the prevalence rates reported by household surveys shift as a result. [2]

Despite its plausibility, this bias has received little attention in the existing literature. Evidence that time-of-day influences body size comes from a handful of clinical studies in high-income settings (Siklar et al., 2005; Voss & Bailey, 1997; Wright et al., 2025), typically with a few dozen children under controlled conditions, which document the physiological patterns but were not designed to assess causality and implications for national nutrition statistics. A separate literature in economics shows that interview timing affects the likelihood of disclosing intimate partner violence (Theiss, 2026), but it concerns subjective and self-reported outcomes, while research on anthropometric data quality has focused on random error and implausible values (Grange et al., 2024; Shireen et al., 2015). To the best of my knowledge, no evidence exists on how the timing of measurement affects the anthropometry of children under five, or the nutrition statistics derived from it in LMICs.

[1] The implications of measurement error for research and undernutrition prevalence depend on whether the error is random or systematic. Random error adds variation around children's true measurements, widening the distribution of z-scores and if the mean of the distribution lies above the minus-two threshold, this added variation tends to increase the share of children classified as undernourished (Grange et al., 2024). It is the kind of error that DHS quality assessments are built to catch, since age heaping, digit preference, implausible z-scores, and inflated standard deviations are its signatures and the observations can be flagged, corrected, or excluded (Shireen et al., 2015). Systematic error is different as it moves every reading in the same direction, shifting the mean and pushing the tails in opposite directions, so that a uniformly overstated weight lowers measured underweight and wasting (Pullum et al., 2020). An error of this kind can enter published estimates and is read as genuine variation across countries or over time.

[2] The same mechanism affects experimental studies. If treated and control children are measured at different times of day, whether because an intervention changes household routines or simply through uneven fieldwork, the time-of-day gradient biases the estimated effect of the intervention on child nutrition.

In this study, I exploit quasi-random variation in the time of day at which children surveyed in DHS conducted in SSA, using the recorded start time of the first household interview as a source of variation in the hour at which children are measured. The identifying assumption is that within a survey cluster and conditional on interview day, child and household characteristics, this timing is independent from children's underlying anthropometric measurement so in the absence of a time-of-day effect, children measured earlier and later should have the same expected weight and height and therefore same nutritional status. The assumption is supported by the fact that DHS enumerators are assigned pre-selected households within sampled clusters on the daily basis with no advance information about their identity or characteristics (ICF, 2024). Therefore, visitation order and hence measurement timing is mainly driven by fieldwork logistics such as cluster layout, distance between dwellings, and respondent availability. The analytical sample is restricted to DHS surveys round from Phase 7 onward and include a sample of 114,996 children under five from 40 surveys in 29 SSA countries, of whom 53.2% were measured standing. [3]

The results show that the time of day predicts whether the child is classified as wasted or underweight but has no effect on stunting. Among children under five, the probability of being wasted or underweight decrease over the day due to increase in weight-based indicator (WHZ and WAZ) so that a child whose household is interviewed after 16:00 is 19 percent less likely to be classified as wasted and 16 percent less likely to be underweight than an otherwise identical child in the same cluster interviewed in the morning during the same day. This result reflects an increase in recorded weight of approximately one percent of mean body weight between morning and evening, an amount consistent with food and fluid intake but sufficient to move some children across the minus two standard deviations threshold to be undernourish. The gradient is concentrated among children aged two to five, and the survey timing has no significant effect on the probability of being classified as stunted nor on HAZ. I conduct a series of tests examining enumerator fatigue, rounding or heaping in recorded measurements, compare hot and cold interview days and none of these tests produces the pattern predicted by the alternative explanations. The gradient is also unchanged under alternative fixed-effect and clustering structures, when interview-day wet-bulb temperature or enumerator fixed effects are included, and when age-in-months fixed effects are used. The pattern also replicates when using the MICS data, where the gradient among children aged more than two years is of the same magnitude as in the DHS.

The estimated timing gradient is large enough to have important implications for both evaluations of child nutrition programs and national estimates of undernutrition. At the mean interview time (6.74 hours after the 06:00 benchmark), the estimates imply that the same child would be recorded 0.03 standard deviations higher in WAZ and 0.04 standard deviations higher in WHZ than at the benchmark. The size of the difference is comparable to the effects commonly reported in evaluations of cash transfers, school-feeding programs, women empowerment on child nutrition. If treatment and control households are measured at systematically different times (for example, because an intervention changes household routines or caregiver availability) part of the estimated program effect may reflect measurement timing rather than a genuine change in nutritional indicators. The same mechanism can lead household surveys to underestimate weight-based undernutrition, as children measured later in the day are recorded as heavier. Re-anchoring measurements to 06:00 increases estimated underweight prevalence by 0.8 percentage points and estimated wasting prevalence by 0.4 percentage points among children under five. Applied to the under-five population, these shifts correspond to a total of roughly 932,000 children who are underweight and 436,000 who are wasted but are recorded as adequately

[3] Household-interview start times began to be recorded in most countries from DHS phase 7 onward.

nourished because of the survey timing. This bias is positive in every country and is concentrated in the most populous countries (including Nigeria, Tanzania and Kenya).

This paper contributes to three strands of development economics. First, it contributes to the literature on the quality of anthropometric data in LMICs. Existing work has focused on non-directional error and implausible values that can be flagged, corrected, or excluded (Grange et al., 2024; Shireen et al., 2015), this study identifies a source of systematic bias that can shifts national estimates of child undernutrition. Second, the findings have implications for the large literature evaluating nutrition-sensitive interventions and other program using child anthropometry as an outcome. Time-of-day effects represent a form of non-classical measurement error in weight-based indicators and may affect estimated program impacts when measurement timing differs between treatment and control groups. Third, it extends research on survey-design effects, which has largely focused on self-reported outcomes, by showing that seemingly minor features of fieldwork can also distort objectively measured indicators (Beegle et al., 2012; Bound et al., 2001; Kilic & Sohnesen, 2015).

The remainder of the paper is organized as follows. Section 2 describes the DHS data and the construction of the timing variable. Section 3 presents the empirical specification, the identification assumption and the tests of that assumption. Section 4 reports the main results and explores heterogeneity. Section 5 examines the mechanisms. Section 6 presents robustness checks, including an independent replication in the MICS. Section 7 develops the implications for research on child nutrition and for national undernutrition statistics.

2. Data and measurement

2.1 The Demographic and Health Surveys (DHS)

The DHS is a program of nationally representative household surveys begun in 1984, funded primarily by the United States Agency for International Development (USAID) and implemented by ICF in partnership with national statistical agencies and a consortium of international organizations.[4] Since its inception the program has supported more than four hundred surveys in over ninety LMICs. Its stated purpose is to produce accurate, nationally representative data on fertility, family planning, maternal and child health, gender, HIV, malaria and nutrition, and to do so in a form that can be used directly for policy formation, program planning and evaluation.

The DHS is one of the single programs which applies one standardized instrument repeatedly across countries and over time. Questionnaires, interviewer and supervisor manuals, biomarker protocols, training materials and equipment specifications are written centrally and revised together in numbered of phases. A standard survey fields at minimum a Household Questionnaire and a Woman's Questionnaire, usually a Man's Questionnaire, and a separate Biomarker Questionnaire in which anthropometry, anemia testing and other measured outcomes are recorded. Because the protocol is common to every survey and the microdata are released in a harmonized recode format, the same question can mean the same thing in Nigeria in 2018 as in Lesotho in 2023.

The program influence on how child health and nutrition are measured is difficult to overstate. Across much of the developing world the DHS is the only recurrent source of nationally representative anthropometric measurements for children under five, so that national prevalence figures for stunting,

[4] USAID is the primary funder, with additional contributions from other donors and from participating countries; ICF has implemented the program since 2013 together with a consortium of international partner organizations.

wasting and underweight are in practice DHS figures.[5] Those estimates feed the international monitoring systems built on top of them, including the joint child malnutrition estimates maintained by UNICEF, WHO and the World Bank and reporting against the Sustainable Development Goals 6. Because the microdata is released publicly and without charge, they have also become a standard input to academic research on early-life health.

2.2 Sample design and the organization of fieldwork

DHS samples are drawn in two stages. First, enumeration areas from the most recent census frame are selected with probability proportional to size within strata; these selected areas constitute the survey clusters. Second, a complete listing of households is conducted in each selected cluster. Every occupied residential household is assigned a serial number, and a fixed number of households is then selected from that list using systematic sampling with a random start. Although the target sample is defined in terms of eligible women rather than households, typically about 25 to 30 households per rural cluster and 20 to 25 per urban cluster, the two are nearly equivalent in practice because the number of women aged 15 to 49 is close to one per household (ICF, 2012).

Two features of this procedure are important for the design of this paper. First, the set of households to be visited is fixed before fieldwork begins and cannot be changed in the field. The sampling manual states that households are pre-selected at headquarters before fieldwork starts, interviewers are instructed to visit only the pre-selected households, and no substitutions or replacements are permitted to prevent bias. Consequently, field teams enter each cluster with a predetermined list of roughly 25 households and cannot replace one household with another after observing its characteristics. Second, interviewers have limited information about a household before entering it. Fieldwork is carried out by teams composed of a supervisor, a field editor, several interviewers, one or more biomarker technicians, and a driver (ICF, 2024). Each morning, the supervisor or field editor distributes the assigned households among interviewers, who record the assignments on a sheet. This sheet contains only basic identifiers, such as the household number, structure number or address, the name of the household head, and an indicator for the male subsample. The manuals also do not prescribe a rule for assigning households to interviewers or for the order in which households are visited.

2.3 Anthropometric measurement

Anthropometric data are collected for all children aged 0–59 months who slept in the sampled household on the night before the interview. Measurements are taken by trained biomarker technicians and recorded in a dedicated Biomarker Questionnaire (ICF, 2021). Children are weighed using a digital scale that records measurements in 0.1-kilogram increments. They are measured in their underwear, or with minimal clothing, to reduce the risk that garments influence the recorded weight. Length or standing height is measured using a standardized measuring board and recorded to the nearest 0.1 centimeters. This procedure requires two trained staff members: one positions the child, ensures that the measurement is taken correctly, and reads the value aloud; the other assists in positioning the child and records the measurement.

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[5] The claim concerns recurrence and national representativeness rather than exclusivity. MICS measures child anthropometry in many of the same countries, and national nutrition and SMART surveys do so intermittently; what distinguishes the DHS is the combination of national representativeness, a standardized protocol and repetition on a regular cycle.

[6] The Joint Child Malnutrition Estimates are produced by UNICEF, WHO and the World Bank, a group "created in 2011 to address the call for harmonized child malnutrition estimates", which "releases annually estimates for child stunting, overweight, underweight, wasting and severe wasting" compiled from nationally representative household surveys (UNICEF, WHO and World Bank, 2023). Stunting and wasting prevalence are the indicators tracked under Sustainable Development Goal target 2.2.

The measurement position depends primarily on the child's age. Children younger than two years are measured lying down, or recumbently, whereas children aged two years and older are measured standing (ICF, 2012). This distinction reflects both practical and measurement considerations. Young children may be unable to stand independently, remain upright, and maintain the required position long enough to produce a reliable measurement. Recumbent measurement therefore provides a more appropriate procedure for this age group. In addition, recumbent length and standing height are distinct anthropometric measures. When measured lying down, the spine is not compressed by the child's body weight, resulting to the recumbent length to be about 0.7 centimeters greater than standing height for the same child. The WHO Child Growth Standards account for this difference by using recumbent length for children below 24 months and standing height for those aged 24 months and older. Where the measurement position differs from the one appropriate for the child's age, an adjustment of approximately 0.7 centimeters is applied (WHO, 2006).

Weight, height and age are the inputs used to construct the three anthropometric z-scores, each of which compares a child's measurement with the distribution observed among children of the same sex (WHO, 2006). HAZ compares a child's height with the expected height of children of the same age and sex. Because linear growth reflects nutritional conditions accumulated over time and past growth deficits are difficult to reverse, this indicator captures chronic undernutrition. WHZ compares a child's weight with the expected weight of children of the same height and sex, thereby measuring thinness relative to body size. Because weight can respond relatively quickly to illness or short-term food shortages, a child whose WHZ falls more than two standard deviations below the reference median is classified as wasted, indicating acute undernutrition. WAZ compares a child's weight with the expected weight of children of the same age and sex and serves as a composite indicator, since low WAZ may reflect short stature, low weight relative to height, or both. A child whose WAZ falls more than two standard deviations below the reference median is classified as underweight.

2.4 The timing variable

The DHS interview begins when an enumerator arrives at a pre-selected household and obtains informed consent. The enumerator then administers the Household Questionnaire, which records the household composition and identifies members eligible for the individual questionnaires and for biomarker collection. The women's questionnaire follows, after which anthropometric measurement is conducted by the biomarker technician (ICF, 2021). The clock time at which this sequence begins is recorded on the questionnaire itself.

Since Phase 7, the DHS has recorded this start time, along with the time the household interview ends and the resulting duration. The start time is entered by the interviewer as the first item of the Household Questionnaire, immediately after consent is obtained and before the household roster is completed. The date of the biomarker visit is recorded separately, but no start time is collected for that visit. As a result, the household interview time is the only clock-time information available in the surveys. Its availability determines the scope of the analysis, because earlier DHS phases did not record interview times and surveys fielded before approximately 2015 therefore contain no comparable timing measure and cannot be used here.

2.5 The study sample

The analysis uses DHS data for SSA drawn from IPUMS-DHS, a harmonized version of the surveys that applies consistent variable names, codes and value labels across countries and rounds (Boyle et al., 2024). All available child records were downloaded and merged with the eight DHS surveys conducted

between 2023 and 2025 producing more that 1,523,149 observations.[7] Children were selected on the basis of three criteria: (i) the survey had to record the start time of the household interview, which confines the sample to 41 surveys;[8] (ii) the child had to fall in the anthropometry subsample and to have actually been measured; and (iii) the record had to carry a valid start time, a valid z-scores and a valid measurement position. The appendix Table 5 shows how each criterion affects the sample size, with all criteria combined leaving a final sample of 263,665 children.

Three additional sample restrictions support the use of household-interview timing as a quasi-random proxy for when children are measured. First, the analysis retains children in households interviewed during the enumerator's first visit. Households interviewed on a second or third visit are excluded as their interview timing may depend on household characteristics, such as employment, income, or preferences, that could also be related to children's nutritional status. Second, anthropometric measurements must have been taken on the same day as the household interview as otherwise the recorded interview time provides no information about measurement timing. Finally, the sample is restricted to households with interview start times between 06:00 and 22:59 to exclude implausible or overnight timestamps.

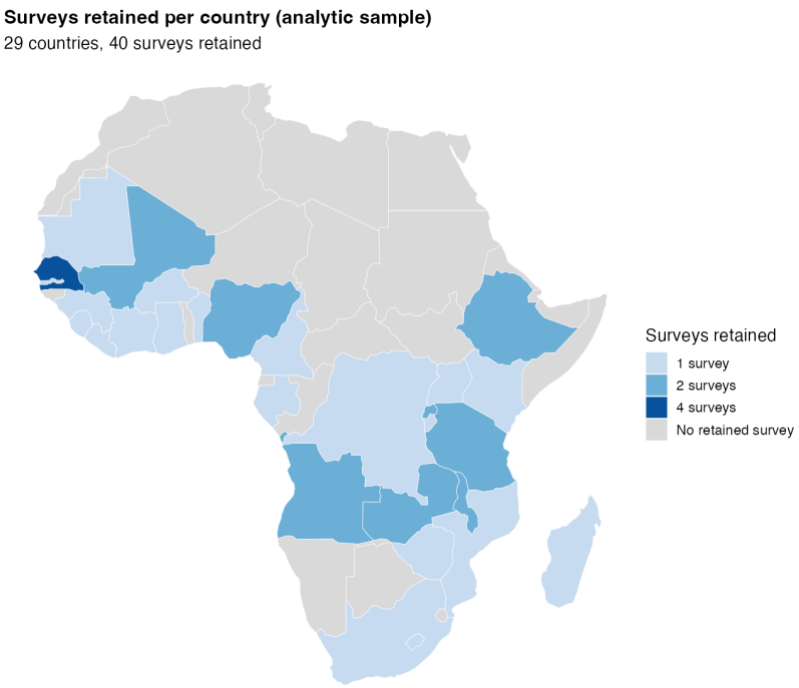
The final sample include 114,996 children from 40 surveys conducted between 2015 and 2025 across 29 countries. Table 6 in the appendix presents the sample distribution by country, with countries listed from the largest to the smallest contribution to the sample and Figure 1 below shows the geographic coverage. Tanzania contributes the largest share of observations, followed by Kenya, Senegal, Nigeria, and Zambia; together, these five countries account for 38 percent of the total sample. Sample size ranges from 345 to 10,200 with twenty countries contribute a single survey round, eight contribute two rounds,[9] and Senegal contributes four.[10] The sample also covers most of West Africa, as well as much of East and Southern Africa and Madagascar. The main gaps are in Central Africa and the eastern Sahel, where Chad, Niger, the Central African Republic, and South Sudan do not meet the selection criteria. Across survey years, 2015, 2018, and 2022 account for 45 percent of observations, while 2020 includes only a single survey conducted in Mauritania, reflecting the interruption of DHS fieldwork during the COVID-19 pandemic. The two latest survey years (2024 and 2025) contribute to 12 percent of the sample.[11] Although identification relies on comparisons within clusters and surveys, the uneven distribution of observations across countries and years affects their relative contributions to the pooled estimate, therefore Section 6 assesses the sensitivity of the results by excluding each country and survey year in turn.

The appendix Table 7 compares the study sample with every other child with valid anthropometry in the extract including the pre-2015 rounds together with the 2015 and later children removed by the restrictions. The comparison group spans 157 surveys in 47 countries and 740,855 children. With samples of this size most differences are statistically significant but they are small in magnitude (all normalized differences remain below the conventional 0.25 threshold) except for the share of mothers

[7] IPUMS-DHS does not harmonize the two variables on which this design depends, the clock time at which the household interview begins and the date of the biomarker visit; both were recovered from the original DHS household recode files and merged onto the extract using the survey, cluster and household identifiers. The analysis therefore combines the harmonized extract, which supplies the anthropometric outcomes and the child and household covariates, with the raw survey files, which supply the timing information.
[8] Since enumerators recorded the start time of the household questionnaire from Phase 7 onward.
[9] The eight countries are Angola, Ethiopia, Malawi, Mali, Nigeria, Rwanda, Tanzania, and Zambia.
[10] Senegal contributes four rounds because it operates a continuous DHS, which fields annual survey rounds rather than a single round every five years; the 2017, 2018 and 2019 rounds are part of that program and the 2023 round is a standard DHS.
[11] I further conduct a robustness check by excluding all surveys conducted after 2020 to avoid bias arising from the COVID-19 shock.

with no schooling and the share of children whose father's education is missing reflecting the more recent nature of the sample with and increase in education rather than the timing design itself. [12]

Figure 1 : Survey coverage across Sub-Saharan Africa



***Notes**: Number of survey rounds retained in the study sample by country. Grey indicates a country with no retained survey, either because it has no DHS round beginning in 2015 or later, or because the round it has does not satisfy the selection criteria.*

2.6 Descriptive statistics.

Interview timing in Panel C provides important variation for the interview timing variable that inform the empirical design. Household interviews begin an average at 11:58, with an interquartile range from 10:00 to 13:42, approximately 9.2 percent begin after 16:00. A typical comparison therefore spans several hours, while the full observed range covers roughly seventeen hours. The distribution is right-skewed and concentrated in the late morning. The three middle within-day quintiles together cover approximately four and a half hours, from 09:35 to 14:12, whereas the earliest quintile spans about three and a half hours and the latest covers nearly nine hours. This unbalanced distribution motivates the use of several treatment measures: (i) the continuous specification that imposes a linear relationship across a period with uneven observation density, (ii) the within-day quantile that contain approximately equal shares of the sample but cover intervals of different duration.

Table 1 reports summary statistics for the study sample in three panels covering child characteristics, household characteristics, and household interview timing. The child sample in Panel A is distributed almost uniformly across the 0-59 month and provides approximately 1,900 observations at each month of age with a mean age is 28.1 months. The average child in the sample measures 83.2 centimeters and weighs 11.20 kilograms. The dispersion is wide, with an interquartile range of 21.5 centimeters in height and 5.2 kilograms in weight. The sample is broadly comparable to the wider region, with stunting, underweight, and wasting prevalences of 27.7, 12.9, and 4.7 percent, respectively. These figures are

[12] Maternal education is the only characteristic that approaches a meaningful difference between the study and eligible samples. Mothers in the study sample have, on average, 1.23 more years of schooling with a normalized difference of 0.27. If time-of-day effects are larger in less educated households because measurement conditions are more constrained, the estimated gradient may overstate the population-wide effect.

close to the regional estimates reported by the Joint Child Malnutrition Estimates suggesting that the sample restrictions generated a broadly representative subset of the DHS surveys.

The household characteristics in Panel B indicate that the sample is relatively poor and predominantly rural. 31 percent of child's mothers have no formal education, 37 percent have primary schooling only and 5 percent have reached higher education. Approximately one-third of children live in urban areas, households contain 7 members on average, and the average household includes just over two children under five. The national wealth-quintile distribution is also concentrated toward the lower end, with 23.1 percent of children in the poorest quintile household compared with 16.8 percent in the richest. This imbalance reflects the composition of the DHS anthropometry subsample, which tends to over-represent rural and poorer households. Mothers are relatively young with an average of 29 years and had almost 4 births.

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Table 1 : Summary statistics, study sample

Variable	Mean	SD	Min	p25	Median	p75	Max	% miss	N
A. Child characteristics									
Age (months)	28.09	17.37	0.0	13.0	27.0	43.0	59.0	0.0	114,996
Female	0.500	0.500						0.0	114,996
Aged 24–59 months	0.532	0.499						0.0	114,996
Birth order	3.37	2.23	1.0	2.0	3.0	5.0	18.0	0.0	114,996
Twin (multiple birth)	0.031	0.172						0.0	114,996
Height (cm)	83.24	14.28	10.4	73.0	84.2	94.5	182.9	0.0	114,996
Weight (kg)	11.20	3.50	0.5	8.6	11.2	13.8	64.9	0.0	114,996
HAZ (height-for-age)	−1.25	1.34	−6.0	−2.1	−1.3	−0.4	6.0	0.0	114,996
WAZ (weight-for-age)	−0.81	1.10	−6.0	−1.5	−0.8	−0.1	5.0	0.1	114,935
WHZ (weight-for-height)	−0.13	1.14	−5.0	−0.9	−0.1	0.6	5.0	0.4	114,497
Stunted (HAZ < −2)	0.277	0.447						0.0	114,996
Wasted (WHZ < −2)	0.047	0.211						0.4	114,497
Underweight (WAZ < −2)	0.129	0.335						0.1	114,935
B. Household characteristics									
Mother's age (years)	29.29	6.91	15.0	24.0	28.0	34.0	49.0	0.0	114,996
Parity	3.66	2.28	1.0	2.0	3.0	5.0	18.0	0.0	114,996
Mother's education									
None	0.310	0.462						0.0	114,996
Primary	0.369	0.482						0.0	114,996

Variable	Mean	SD	Min	p25	Median	p75	Max	% miss	N
Secondary	0.271	0.444						0.0	114,996
Higher	0.051	0.220						0.0	114,996
Father's education									
None	0.241	0.428						0.0	114,996
Primary	0.172	0.377						0.0	114,996
Secondary	0.307	0.461						0.0	114,996
Higher	0.074	0.262						0.0	114,996
Urban	0.324	0.468						0.0	114,996
Household size	6.98	4.40	2.0	4.0	6.0	8.0	62.0	0.0	114,996
No. children < 5	2.05	1.30	0.0	1.0	2.0	2.0	20.0	0.0	114,996
Head's age	40.72	13.55	15.0	31.0	38.0	47.0	95.0	0.3	114,697
Female-headed household	0.239	0.427						15.2	97,521
HH interview length (min)	26.50	15.19	0.0	16.0	23.0	32.0	97.0	0.0	114,996
Wealth quintile									
Poorest	0.231	0.422						0.1	114,848
Second	0.213	0.410						0.1	114,848
Third	0.197	0.398						0.1	114,848
Fourth	0.191	0.393						0.1	114,848
Richest	0.168	0.374						0.1	114,848
C. Household interview time									
Hours since 06:00	5.97	2.67	0.0	4.0	5.5	7.7	17.0	0.0	114,996
Start hour (clock, decimal)	11.97	2.67	6.0	10.0	11.5	13.7	23.0	0.0	114,996
Within-day tertile									
T1 (06:00–10:24)	0.333	0.471						0.0	114,996
T2 (10:24–12:44)	0.333	0.471						0.0	114,996
T3 (12:44–22:59)	0.333	0.471						0.0	114,996
Within-day quintile									
Q1 (06:00–09:34)	0.200	0.400						0.0	114,996
Q2 (09:34–10:49)	0.200	0.400						0.0	114,996
Q3 (10:49–12:10)	0.200	0.400						0.0	114,996
Q4 (12:10–14:12)	0.200	0.400						0.0	114,996
Q5 (14:12–22:59)	0.200	0.400						0.0	114,996

***Notes**: Study sample, N = 114,996 children. All statistics are survey weighted. Share variables report the mean, which is the proportion, and leave the quantile columns blank. Undernutrition uses the standard WHO cut-offs. Parental education is shown as the categories in which it enters the regressions; the father's education is not recorded for a fifth of children and carries an explicit missing category. Within-day tertile and quintile ranges are empirical, cut on the pooled distribution of household start times in the study sample. No imputation is performed: "% miss" is the listwise-deletion share each variable would cost, and N is the unweighted count of children contributing to that row.*

3. Empirical strategy

In this section I presented the empirical strategy to measure the causal effect of the hour of measurement on a child's recorded anthropometry. It sets out the estimating equation, the assumption under which its

coefficient can be given a causal reading and reports the tests through which that assumption is examined.

3.1 Estimating equation

The main empirical specification is:

$$Y_{ihcd} = \beta_0 + \beta_1 Hours_{hc} + \gamma' X_{ihc} + \theta_c + \delta_d + \varepsilon_{ihcd} \qquad (1)$$

Y_{ihcd} is the recorded outcome for child i, living in household h, in survey cluster c, and interviewed on day d. The analysis considers several outcomes including the weight in kilograms, height in centimeters, three anthropometric z-scores (HAZ, WAZ, and WHZ), and three binary indicators for stunting, underweight, and wasting. $Hours_{hc}$ is a continuous measure of interview timing defined as the number of hours elapsed since 06:00am when the enumerator began the household questionnaire. The coefficient of interest is β_1 and it captures the change in the recorded outcome associated with an additional hour of the fieldwork day. The term θ_c denotes a full set of DHS cluster fixed effects. Including cluster fixed effects means that the estimates come entirely from differences in interview timing among households within the same cluster and fieldwork round.[13] The term δ_d denotes day-of-week fixed effects, which capture differences in fieldwork patterns across days of the week. X_{ihc} is a vector of child and household characteristics that are predetermined with respect to interview timing. These include the child's sex, age in months and its square, birth order, and twin status; the mother's age, parity, and level of education; the father's education; and household wealth quintile, urban status, household size, number of children under five, and the age, sex of the household head and the length of household module in minutes.[14] All estimates use DHS sampling weights and standard errors are clustered at the survey-cluster level.[15]

3.2 Identification assumption

The primary identification assumption underlying equation (1) is that the start time of the household questionnaire is as good as randomly assigned among children measured during an enumerator's first visit, conditional on the survey cluster, the day of the week, and predetermined child and household characteristics. Under this assumption, β_1, captures the causal effect of measurement timing on recorded anthropometry rather than differences in which children happen to be visited at different hours.

This assumption follows from the DHS sampling and fieldwork protocol as stated in Theiss, (2026). Households are selected before fieldwork begins and cannot be replaced in the field. Supervisors assign pre-selected households to enumerators and no substitutions are permitted (ICF, 2012). As a result, teams cannot decide in advance which households to reach at which hour. The protocol also specifies that enumerators are not allowed to know the members of the households they are assigned to interview. For that reason, the start time of the household questionnaire during the first visit provides the most credible source of quasi-random variation in timing. At that point, the enumerator has no access to the household roster or to information on the demographic or socioeconomic composition of the household, so the first visit cannot be scheduled strategically based on characteristics correlated with child

[13] Survey and region fixed effects are nested within the cluster and absorbed by it. The same applies to every cluster-level characteristic, observed or not: urban status, terrain, distance to a market, nightlights, population density.
[14] Father's education and the sex of the household head carry an explicit "Missing" level for 15.2 percent of children. Observations with a missing continuous control are dropped listwise.
[15] Because the hour of the visit can be summarized in more than one way, and visits are unevenly distributed across the day, I estimate several other versions as a robustness check in Section 6.

anthropometric indicator. If an enumerator's first attempt to conduct an interview is not successful, household members may arrange a callback, and the hour recorded on a completed callback reflects household availability and therefore is less likely to be exogeneous. The possibility of appointment-making is limited before the first attempt. In villages, supervisors may notify local leaders that a team will be working in the cluster, and those leaders may inform the households to expect a visit but there is no standardized process by which a household can pre-schedule the hour of the first visit. The interviewer's manual makes clear that supervisors assign enumerators to make first contact with each household, and that first contact occurs when the enumerator arrives at the dwelling, introduces themselves, and obtains informed consent (ICF, 2024).

While I cannot directly observe whether informal planning of visit times occurs before the first attempt, three considerations suggest that this is unlikely to bias the estimates. First, if local leaders were recommending visit times, those recommendations would need to be correlated with the anthropometric indicators and undernutrition status of the children in the household, which is unlikely as it cannot be assessed without direct measurement. Second, household self-selection is unlikely because households would need to know both the exact hour of the enumerator's visit and that child weight is the outcome of interest, but such information is not provided by the protocol. Third, the estimates are robust to the inclusion and exclusion of a wide range of child and household controls, implying that any selection operating through observables would leave the coefficient essentially unchanged.

3.3 Testing the identification assumption

If visit hour is as good as randomly assigned, predetermined characteristics should not correlate with it. The balance test uses the main estimating equation, with each variable of the specification taken in turn as the dependent variable and hours since 06:00 as the regressor and estimate the following equation:

$$Z_{ihcd} = \varphi_0 + \varphi_1 Hours_{hc} + \gamma' X_{ihc}^{-Z} + \theta_c + \delta_d + u_{ihcd} \qquad (2)$$

where Z is the variable being tested and X^{-Z} is the control vector with the dependent variable, and any other variable derived from the same underlying source, removed. The fixed effects, clustering, and weighting are the same as in Equation (1).

Table 2 reports one row per coefficient from the main specification, and the omitted category for each categorical variable correctly has no row. Of the twenty-four predetermined terms, only the household size survives false-discovery-rate correction, which falls by 0.032 persons per hour, so that households reached at the end of a twelve-hour fieldwork day contain about four tenths of a person fewer than those reached at its start. In standardized terms, household size moves by 0.007 standard deviations of its own dependent variable per hour so that a ten-hour difference in visit time therefore shifts the average control under a tenth of a standard deviation. The child's sex, twin status, the mother's age, and the number of children under five are all statistically not different from zero. Conditional on the cluster and weekday fixed effects and the control set, the hour at which a household is reached is therefore exogeneous.

Table 2 : Balance: every main-specification term regressed on hours since 06:00

Main-specification term	Hours since 06:00	FDR q	SD of term	Coef / SD
Child female	0.000 (0.001)	1.000	0.500	0.000
Age (months)	0.020 (0.027)	0.952	17.385	0.001
Age (months) squared	1.181 (1.582)	0.952	1039.697	0.001
Birth order	0.001 (0.001)	0.952	2.260	0.000
Single birth (vs multiple)	0.000 (0.000)	1.000	0.171	0.000
Mother's age	0.009 (0.009)	0.952	6.963	0.001
Parity	−0.001 (0.001)	0.952	2.305	−0.000

Main-specification term	Hours since 06:00	FDR q	SD of term	Coef / SD
Mother's education: primary	−0.000 (0.001)	1.000	0.481	−0.000
Mother's education: secondary	−0.001 (0.001)	0.952	0.443	−0.002
Mother's education: higher	0.001** (0.000)	0.226	0.204	0.005
Father's education: primary	−0.001 (0.001)	0.952	0.376	−0.003
Father's education: secondary	−0.001 (0.001)	0.952	0.458	−0.002
Father's education: higher	0.001 (0.001)	0.583	0.253	0.004
Wealth quintile: second	0.001 (0.001)	0.660	0.412	0.002
Wealth quintile: third	−0.002** (0.001)	0.110	0.398	−0.005
Wealth quintile: fourth	−0.001 (0.001)	0.617	0.383	−0.003
Wealth quintile: richest	0.000 (0.001)	1.000	0.351	0.000
Urban	0.000 (0.000)	1.000	0.462	0.000
Household size	−0.032*** (0.005)	0.001	4.380	−0.007
No. children < 5	0.001 (0.002)	1.000	1.289	0.001
Head's age	−0.057** (0.024)	0.110	13.686	−0.004
Female-headed household	−0.000 (0.001)	1.000	0.405	−0.000
Household interview length (minutes)	−0.214*** (0.022)	0.001	15.241	−0.014

***Notes**: Each row is a separate regression of that term of the main specification on hours since 06:00, under the full control set, with DHS cluster and day-of-week fixed effects and standard errors clustered at the cluster level, estimated on the sample of 114,550 children complete on every control. Factor levels are expanded exactly as they enter the outcome regression, so the rows are the coefficients of the estimating equation, and the omitted category of each factor (boys, singletons, mothers and fathers with no schooling, the poorest wealth quintile, male-headed households) has no row. Every term built from the same base variable is removed from the right-hand side when that variable is the dependent variable. Urban residence is constant within a DHS cluster and absorbed by the cluster fixed effects, so its row is identically zero. "Coef / SD" expresses the coefficient in standard deviations of the dependent variable, whose standard deviation is given in the previous column. Anderson (2008) sharpened q-values are computed within the family of twenty-five dependent variables. Survey weights applied. *** p < 0.01, ** p < 0.05, * p < 0.1.*

3.4 Selection into measurement

The balance test is conditional on children who were measured and therefore does not rule out selection into measurement. Anthropometry is collected in a subsample of households and for children who are present and whose caregiver provides consent. If teams arriving later in the day were less likely to complete the anthropometric module, or if the children measured at different times differed systematically from those not measured, the analytical sample would be selected on the treatment itself. In that case, balance among measured children would not reveal the resulting bias.

To Address this concern, I construct a child-level dataset containing all 473,738 children identified in the 41 Phase 7 and onward surveys with information on household interview start time within the valid timing window (between 6:00 to 23:00) regardless of whether they were finally measured or not. Anthropometric data are available for about 60 percent of these children. I then estimate the following linear regression model:

$$Measured_{ihcd} = \pi_0 + \pi_1 Hours_{hc} + \gamma' W_{ihc} + \theta_c + \delta_d + v_{ihcd} \qquad (3)$$

Where $Measured_{ihcd}$ takes a value of one if the child i in household h, cluster c, and interview day d was measured and zero otherwise. W_{ihc} is a vector of predetermined characteristics, θ_c is the cluster fixed effects and δ_d is the survey date fixed effects. The clustering structure are the same as those used in Equation (1).

Table 3 reports the coefficients from this model in two panels. In Panel A, the coefficient on visit hour is positive but not statistically significant suggesting that the survey timing is not related to the probability of measurement. Panel B shows that the model has sufficient power to detect selection when it is present. The coefficients on the control variables show that girls, children of higher birth order, whose father has secondary education and living in female-headed households are more likely to be measured, whereas children of higher-parity mothers and children whose father's education is not recorded are less likely to be measured. The results therefore provide no evidence that the visit hour selects children into the anthropometry sample.

Table 3 : Selection into measurement: the visit hour and the controls

Panel	Term	Estimate (SE)	Standardized coef.
A. Treatment	Hours since 06:00	0.002 (0.002)	0.015
B. Controls	Child female	0.008** (0.004)	0.008
	Birth order	0.052*** (0.003)	0.245
	Mother's age	0.000 (0.001)	0.005
	Parity	−0.048*** (0.004)	−0.231
	Mother's education (ref. none)		
	Primary	−0.009 (0.007)	−0.008
	Secondary	−0.002 (0.008)	−0.002
	Higher	−0.003 (0.017)	−0.001
	Father's education (ref. none)		
	Primary	0.008 (0.009)	0.006
	Secondary	0.020** (0.008)	0.018
	Higher	0.022 (0.014)	0.012
	Wealth quintile (ref. poorest)		
	Second	0.006 (0.008)	0.005
	Third	0.007 (0.008)	0.006
	Fourth	0.003 (0.009)	0.002
	Richest	0.009 (0.012)	0.007
	Head's age	0.000 (0.000)	0.005
	Female-headed household	0.022*** (0.008)	0.018
Observations		445,721	

Notes: *One linear probability model; the dependent variable equals one if the child was measured. The universe is all 473,738 children in the 41 Phase 7 and onward surveys with a valid household start time inside the 06:00–22:59 window, measured or not, with no restriction on first visit, same day or valid anthropometry; the regression uses the 445,721 of them complete on every control, of whom 60 percent were measured. Panel A is the treatment and Panel B the controls, entered exactly as in Equation (3). Omitted categories are boys, mothers and fathers with no education, the poorest wealth quintile and male-headed households; urban residence and a missing head's sex are constant within cluster and absorbed by the cluster fixed effects, so they have no row. The child's age in months and its square are excluded because they are recorded on the anthropometry record and are missing for every unmeasured child. The household interview length is excluded because it is realised during the visit. The standardized coefficient is the coefficient multiplied by the standard deviation of its regressor and divided by the standard deviation of the dependent variable, so that rows are comparable; the stars give the significance level. DHS cluster and day-of-week fixed effects; standard errors clustered at the cluster level; survey weights applied. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.*

4. Results

This section reports the effect of the hour of measurement on the probability for a child to be classify as wasted, underweight or stunted, and on the three z-scores. The treatment throughout is the number of hours elapsed between 06:00 and the start of the household interview.

4.1 Main findings

Table 4 reports the coefficient on hours since 06:00 for eight outcomes across three panels. The weight-based block includes the probabilities of being classified as wasted or underweight, the WHZ and WAZ scores from which these classifications are derived, and recorded weight in kilograms. The height-based block includes the probability of being classified as stunted, the HAZ score, and recorded height in centimeters. All columns use the full control set, together with cluster and day-of-week fixed effects.

It shows that each additional hour reduces the probability of being classified as wasted by 0.076 percentage points and as underweight by 0.109 percentage points. These changes are driven by increases of 0.006 in WHZ and 0.005 in WAZ per hour, which in turn reflect an increase in recorded weight of 0.010 kilograms per hour. A ten-hour difference in measurement time therefore corresponds to an increase of about 0.1 kilograms, or one percent of the sample mean weight of 11.17 kilograms. The height-based block is flat and is not significant effect. The height-based block is null as none of the stunted, HAZ and height is statistically different from zero. Stunting, which is cut from HAZ, is therefore unaffected by the hour at which the household was interviewed, and weight and the indicators built from it are the only outcomes that respond to the hour of measurement.

Table 4 : Effect of the interview hour on recorded anthropometry

	Weight-based outcomes					Height-based outcomes		
Treatment	**(1) Wasted (pp)**	**(2) WHZ**	**(3) Underweight (pp)**	**(4) WAZ**	**(5) Weight (kg)**	**(6) Stunted (pp)**	**(7) HAZ**	**(8) Height (cm)**
Hours since 06:00	−0.076** (0.038) [0.065]	0.006*** (0.002) [0.018]	−0.109* (0.063) [0.089]	0.005** (0.002) [0.042]	0.010*** (0.003) [0.008]	−0.005 (0.076) [0.467]	0.002 (0.002) [0.217]	0.010 (0.008) [0.150]
Mean of outcome	4.961	−0.163	13.289	−0.826	11.168	27.754	−1.251	83.216
R^2	0.203	0.293	0.243	0.332	0.857	0.293	0.350	0.927
Observations	114,052	114,052	114,489	114,489	114,550	114,550	114,550	114,550

***Notes**: All children aged 0–59 months of the analytical sample. Each cell is the coefficient on hours since 06:00 from a separate regression of the outcome named in the column heading, with the standard error in parentheses. The full control set is the child's sex, age in months and its square, birth order and twin status; the mother's age, parity and level of education; the father's education; the household's wealth quintile, urban residence, size and number of children under five; the age and sex of the household head; and the length of the household interview module. Every regression includes DHS cluster and day-of-week fixed effects, uses the DHS child weights, and clusters standard errors at the DHS cluster. Wasted, underweight and stunted are linear probability models for the indicator that the z-score falls below −2, with the coefficient scaled to percentage points. Values in square brackets are Anderson (2008) sharpened false-discovery-rate q-values, computed within the family of the eight outcomes. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.*

4.2 Heterogeneity

This subsection explores whether the timing gradient differs across children and households. Appendix Table 8 reports, for each of seven moderators, the hours-since-06:00 gradient in the base group, the interaction term, and the marginal effect implied for the other group with its own standard error. [16] Of the fifty-six interactions estimated across the seven moderators and eight outcomes, six are significant at the five percent level, about what chance alone would produce, and none survives the sharpened false-discovery-rate correction.

The moderator that matters most for reading the paper is age. Panel E interacts the gradient with an indicator for being aged 24 to 59 months which correspond to the age limit between measurement standing and lying down. The weight gradient is 0.003 kilograms per hour among children under two and 0.015 among children aged 24 to 59 months and the difference is statistically significant. The probability of being classified wasted and the WHZ and WAZ are not different between the two groups. The gradient in recorded weight is therefore driven by the older children which is consistent with a child who eats and drinks over the day since the two groups are measured by the same teams on the same days. The height-based outcomes do not differ systematically between the two groups.

The remaining moderators do not reveal statistically significant heterogeneity. The weight gradient is present among boys and girls (0.007 and 0.013 kilograms per hour), later-born and first-born children (0.011 and 0.008), children from the poorest two wealth quintiles and the richer three (0.009 and 0.012), rural and urban children (0.006 and 0.019), and at the 25th and 75th percentiles of age (0.005 at 13 months and 0.015 at 43 months) but the interactions comparing these groups are individually not different from zero after correction. Children from multiple births, 3,473 in the sample, show no gradient of their own, but the interaction with twin status is small and imprecise. The gradient is therefore a property of the measurement rather than of any group of children with age the one dimension along which it clearly varies.

5. Mechanisms

The results so far establish that recorded weight increase over the fieldwork day while recorded height does not, and that the gradient is concentrated among children aged two to five. This section elaborates the mechanism that explains the findings.

5.1 Child feeding

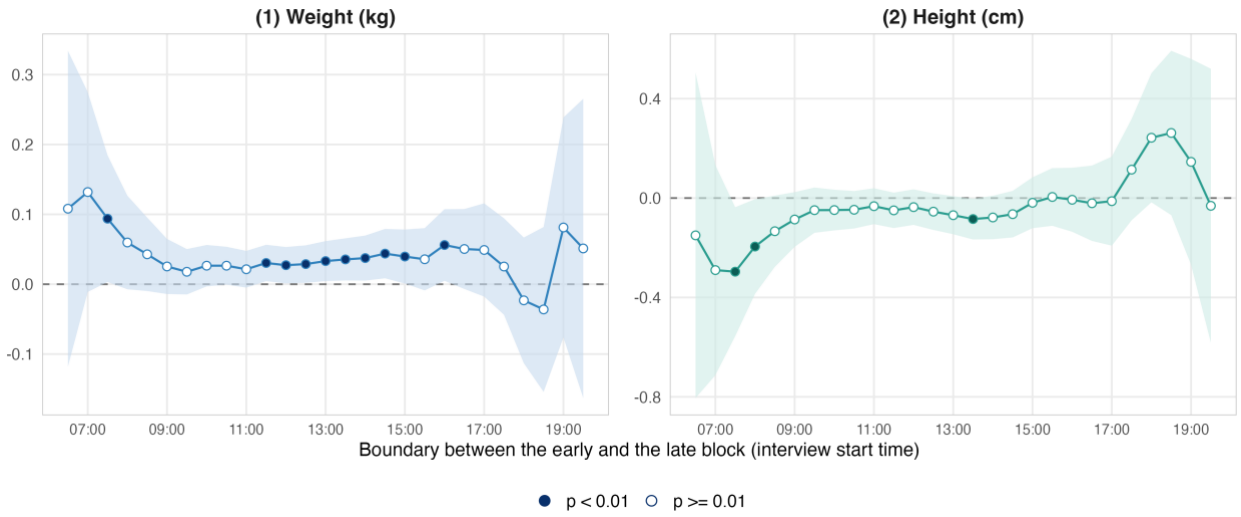
If the observed within-day gradient is consistent with the accumulation of food and fluid intake after waking, three empirical patterns should be observed. First, weight should increase gradually with time elapsed during the day as food intake accumulates through meals, snacks, and fluids while gastric contents are continuously processed. Second, height should remain unchanged, since food and fluid intake increase body mass but not stature. Third, the gradient should be evident from the morning onward. In contrast, a discrete meal-time effect would produce changes concentrated around midday and evening, while a general measurement artifact would not necessarily be confined to weight-related outcomes. These implications are assessed by cutting the fieldwork day into an early and a late block

[16] The interaction terms are treated as one family for multiple-testing purposes.

and moving the boundary between them in half-hour steps which traces how the difference between children measured later and earlier builds up over the day.[17]

Figure 2 bellow reports adjusted differences in recorded weight and height between children interviewed before and at or after half-hour thresholds from 06:30 to 19:30. The weight difference is consistently positive and precisely estimated from 08:00, rising from approximately 0.02 kg at 09:30–11:00 to 0.055 kg at 16:00. Estimates are significant from 11:30 to 16:00 but less precise at the earliest and latest thresholds, where sample sizes are smaller. This pattern suggests gradual weight accumulation throughout the day rather than a discrete increase at mealtimes. Height differences remain within 0.1 cm of zero from 09:00 to 17:00, with isolated significant estimates and greater uncertainty at later thresholds. Figure 3 confirms this pattern using half-hour interview indicators relative to the 06:00 hour. Weight differences increase from 0.08–0.15 kg in the morning to 0.17 kg at 14:30–15:00, 0.18–0.19 kg at 16:00–17:00, and 0.25 kg at 17:30. Estimates from 14:30 onward are significant until sparse late observations reduce precision. Height shows no significant differences or systematic trend. Appendix Figure 6 similarly shows WHZ increasing from approximately 0.15 standard deviations above the reference hour in the morning to 0.25 at 17:00 while height and HAZ exhibit no systematic hourly pattern.

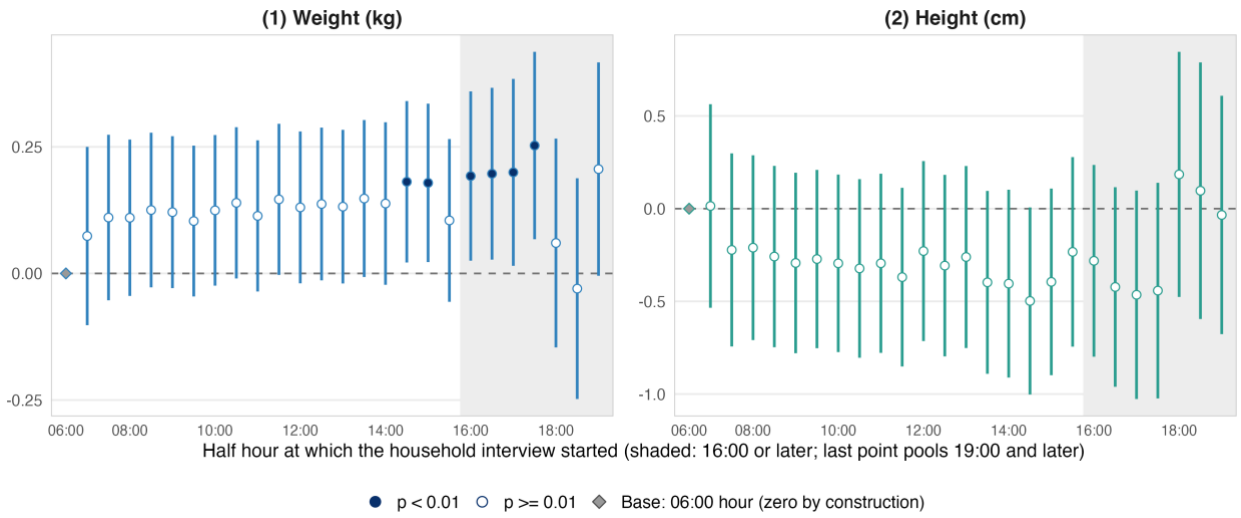
Figure 2 : Recorded weight and height of children interviewed after versus before a moving boundary of the fieldwork day, all children aged 0–59 months



***Notes**: All children aged 0–59 months, full-control specification. The fieldwork day is cut into two blocks, interviews started before the boundary and interviews started at or after it, and the boundary moves from 06:30 to 19:30 in half-hour steps; each point is the coefficient on the late block, relative to the early block, from a separate regression of recorded weight (panel 1) or height (panel 2) with the full control set, DHS cluster and day-of-week fixed effects and survey weights. Shaded bands are 99 percent confidence intervals from standard errors clustered at the DHS cluster; filled points are significant at the one percent level. The earliest and latest boundaries leave only a few hundred children on one side and are correspondingly imprecise. The 16:00 boundary corresponds to the evening block of the robustness tables.*

[17] Moreover, the data do not contain measures of meals, snacks, or infant and young child feeding practices, preventing a direct test of whether the gradient varies with observed feeding behavior.

Figure 3 : Recorded weight and height by the half hour at which the household interview started, all children aged 0–59 months



***Notes**: All children aged 0–59 months, full-control specification. One indicator for each half hour in which the household interview started, with the 06:00 hour omitted as the base; interviews started at 19:00 or later are pooled into the last point. Each point is the coefficient on that half hour from a separate regression of recorded weight (panel 1) or height (panel 2) with the full control set, DHS cluster and day-of-week fixed effects and survey weights; vertical bars are 99 percent confidence intervals from standard errors clustered at the DHS cluster, and filled points are significant at the one percent level. The base hour holds only a few hundred children, so every contrast inherits its imprecision. The shaded band marks interviews started at 16:00 or later.*

5.2 Child clothing

Clothing provides another possible explanation for the observed gradient. If children measured later in the day wear more clothing, and if enumerators become less diligent about removing it, the estimated gradient could reflect clothing instead of changes in the child's weight. This mechanism makes a distinct prediction: since clothing weighs less on hotter days, a clothing-based gradient should weaken as temperature rises. Interacting hours since 06:00 with the interview-day wet-bulb temperature therefore provides a test that distinguishes the two explanations.[18] Appendix Table 9 reports the results. In Panel A, none of the interactions of hours since 06:00 with wet-bulb temperature is statistically significant for any of the eight outcomes. The weight gradient implied at the 25th percentile of temperature (14.9 °C) is 0.013 kilograms per hour and at the 75th percentile (22.4 °C) 0.007, a difference that is within sampling error, while the WHZ gradient is 0.007 and 0.006 and the wasting gradient −0.063 and −0.079 points per hour. Panel B splits the sample instead into interview days hotter and cooler than the cluster's climatological normal: the weight gradient on cool days and that on hot days are not statistically different and the wasting gradient is the same on the two kinds of day. The weight gradient does not decline on hotter days as a clothing-based explanation would predict, and the classification effect does not decline. This result indicates that the weight–height asymmetry is not the reflect of clothing.

5.3 Digit heaping

To assess whether the observed weight gradient could reflect declining measurement quality later in the day, instead of changes in children's body weight, Appendix Table 10 re-estimates the main specification while controlling for an indicator for weights recorded in whole units. Tired or hurried enumerators may be more likely to round measurements, for example, recording whole kilograms rather than decimals and if such rounding increased over the day and were systematically upward, it could create a spurious gradient. However, including the heaping indicator leaves the estimated weight, WAZ

[18] Because all specifications include cluster fixed effects, the interaction is identified from day-to-day variation within clusters.

and WHZ gradient unchanged. The heaping indicator itself is small, significant only for wasting and WHZ, indicating that while rounding occurs in the data, it does not account for the observed time-of-day pattern.

5.4 Enumerator fatigue

A fatigue-based explanation would imply that enumerators measure children less accurately as the day run, and this can create bias in children's anthropometric outcomes. If this is the case then measurement error (flagged, implausible, or missing observations) should become more frequent as the day run. The squared residuals do not increase with interview hour, and visit timing does not predict flagged z-scores, implausible observations, or missing height and weight measurements among the children in the anthropometry module; all estimated coefficients are effectively zero as depicted in Appendix Table 11. Taken together, these results provide no evidence that measurement quality deteriorates during the day, suggesting that the estimated gradient most likely reflect changes in the quantity measured.

6. Robustness

This section subjects the main result to a sequence of checks. Every check re-estimates the hours-since-06:00 specification on the full analytical sample of children aged 0 to 59 months for each outcome. The tables are in the appendix.

6.1 Fixed effects, weighting and controls

The first set of checks examines whether the results are an artefact of how the empirical specification is constructed. Appendix Table 12 reports each test in a separate panel.

6.1.1 Survey weights

Removing the survey weights reduces the estimated gradient for recorded weight, WAZ and the WHZ as depicted in Panel A of Table 12 but the results remain essentially unchanged. All three coefficients are statistically significant at the one percent level, while the height gradient remains statistically not significant. The results are therefore not driven by weighted observations, and the null finding for height is not a result of the weighting procedure.

6.1.2 Age controls

Age receives particular attention as measurement position is determined largely by age. In the baseline specification, age enters as a linear term and its square. Replacing these terms with a complete set of age-in-month fixed effects absorbs the age profile non-parametrically but leaves the estimated coefficients essentially unchanged, as shown in Panel B of Table 12. The timing effect is therefore not explained by differences in the age composition of children measured at different times of day, and the null results for height and HAZ remain.

6.1.3 Seasonality

Seasonality provides another potential explanation. Fieldwork takes place in different months across surveys, and child nutrition may vary over the agricultural year. A time-of-day gradient could therefore reflect the season in which later interviews happen to occur. Adding country-by-month-of-year fixed effects to the cluster fixed effects to control for this effect leaves all five estimates essentially unchanged as illustrated in Panel C of Table 12. This is not surprising as fieldwork in a DHS cluster is generally

completed within a few days, so observations from a given cluster fall generally within the same calendar month.

6.1.4 The physical environment

The physical environment provides a different potential confounder. Later interviews often take place under hotter conditions, and heat may affect both hydration and the amount of clothing a child wears during weighing. Controlling for wet-bulb temperature on the interview day leaves the estimated gradients for weight, WAZ, and WHZ largely unchanged as shown in Panel D of Table 12. All three remain statistically significant at the one percent level. The corresponding coefficients for height and HAZ remain statistically not significant.[19]

6.1.5 Post-treatment controls

Two additional controls are deliberately excluded from the main specification because they are realized after the household visit begins. The length of the women's questionnaire and the total duration of the household interview are outcomes of the visit as well as determinants of the time at which anthropometry is recorded, therefore conditioning on them could affect the results. However, adding the women's questionnaire length (as shown in Panel E of Table 12) or total interview duration (as shown in Panel F of Table 12) as control leave the main results broadly the same. Height and HAZ remain statistically not significant under both specifications. The main specification therefore does not require these post-treatment controls, and their inclusion does not change the conclusions for any of the five outcomes.

6.2 Sample definition

The second set of checks examines whether the results are an artefact of how the analytical sample is constructed.

6.2.1 The same-day measurement restriction

The analytical sample includes children whose anthropometric measurements were collected on the same day as the household interview. This restriction is necessary as the interview start time is informative about the timing of the child's measurement only when both occur on the same day. Here, I relax this restriction to test whether the estimated gradient remains. As reported in Appendix Table 13, expanding the sample by including children measured on any later day, while controlling linearly for the delay, or adding explicit delay indicators and the number of visits as control attenuates the results for the weight and all weight-based outcomes while height and HAZ remain statistically not significant. This attenuation is expected because relaxing the same-day restriction introduces measurement error into the treatment variable. For a child measured two or more days after the household interview, the recorded interview hour is not necessarily correlated to the hour at which the anthropometric measurement occurred.

6.2.2 The COVID-19 interruption

Fieldwork conditions changed during the COVID-19 pandemic. The DHS Program suspended field data collection in April 2020 and surveys resumed at staggered dates thereafter, so rounds fielded on

[19] This specification reduces the sample by approximately 3 percent because of missing temperature data, which explains why its coefficients differ slightly from those in the baseline model.

either side of the interruption were not conducted under identical conditions.[20] Panel A of Appendix Table 14 restricts the sample to the 19 surveys completed before the pandemic was declared, which contain 56,482 of the 114,550 children (49 percent). The weight gradient is positive and significant, and the WHZ, WAZ, wasting and underweight coefficients keep their sign and size but lose significance with half the sample. The results therefore have the same structure in the pre-pandemic subsample as in the full sample, which suggests that the main findings are not generated by changes in fieldwork procedures after March 2020. Panel B drops instead the two surveys, Mali 2023 and Ethiopia 2024, whose measurement position does not follow the 24-month protocol; the coefficients are essentially unchanged.

6.2.3 The start of the fieldwork day

A final sample concern relates to the beginning of the fieldwork day. The earliest interviews are relatively few, raising the possibility that they disproportionately determine the estimated gradient. Panels C and D of Appendix Table 14 re-anchor the treatment at 07:00 and at 08:00 and restrict the sample to interviews beginning at or after the new anchor.[21] The gradients are essentially unchanged for weight WHZ and WAZ under both anchors, and a fall of 0.069 and 0.065 points per hour in the probability of being classified wasted and of 0.102 and 0.083 in that of being classified underweight. Stunting, HAZ and height remain statistically not significant. The timing gradient is therefore not driven by the earliest interviews.

6.3 Alternative definitions of the treatment

The continuous specification assumes a linear relationship between recorded anthropometry and interview time. To relax this assumption, I compare blocks of the fieldwork day using three definitions: fixed morning, afternoon and evening periods; equal-size groups based on the median, tertiles and quintiles of interview start times; and equal-duration intervals spanning 06:00–22:58. Appendix Table 15 to Table 17 report these specifications, while Appendix Table 18 compares the latest and earliest blocks across ten partitions, including alternative evening thresholds. Appendix Figure 6 presents hourly estimates, and Figure 2 in Section 5 varies the early–late boundary in half-hour increments.

The fixed clock cuts indicate that differences are concentrated in the evening**.** Afternoon visits do not differ significantly from morning visits for any outcome. Evening visits, however, are associated with reductions of 0.95 percentage points in wasting and 2.11 points in underweight which represent 19% and 16% of their respective prevalence and increases of 0.052 in WHZ, 0.071 in WAZ and 0.124 kg in recorded weight. All five estimates remain significant after false-discovery-rate correction. The underweight contrast, which the linear specification estimates less precisely, reflects a largely flat profile through mid-afternoon followed by a decline after 16:00. Evening visits also show lower stunting, higher HAZ and greater height.

Equal-size partitions yield smaller but consistent weight differences. Relative to the earliest group, recorded weight is 0.038 kg higher above the median, 0.047 kg higher in the last tertile and 0.060 kg

[20] The DHS Program announced on 9 April 2020 that it had paused in-person trainings, pre-tests and data collection, and announced later in 2020 that some surveys had returned to the field while others were postponed to 2021 (The DHS program, 2020). The program did not publish a protocol describing how fieldwork was modified on resumption, so the restriction here is deliberately. It removes every survey whose fieldwork was not complete before the interruption rather than attempting to identify which procedures changed.

[21] Re-anchoring the treatment on its own cannot change a slope as subtracting a constant from a continuous regressor shifts an intercept that the fixed effects already absorb. The two rows therefore also restrict the sample to interviews beginning at or after the new anchor, which is the substantive version of the question.

higher in the last quintile, all significant at the 5% level or better. Corresponding WHZ differences are 0.020, 0.032 and 0.042, while wasting declines by 0.16, 0.46 and 0.50 percentage points, respectively, significantly only for tertiles. Intermediate groups and height-based outcomes show no significant differences. These smaller contrasts reflect the pooling of relatively flat midday measurements with later observations.

Equal-duration partitions show similar patterns but become less precise as late blocks shrink. Comparing interviews before and after 14:29 yields reductions of 0.47 percentage points in wasting and 0.86 points in underweight, alongside increases of 0.031 in WHZ, 0.027 in WAZ and 0.058 kg in weight, with no height-based differences. Under fifths, the 16:11–19:34 block shows larger contrasts with wasting and underweight decline by 0.97 and 1.79 points, while weight increases by 0.127 kg.

Across the ten partitions, the latest-block weight contrast is always positive, and joint tests reject zero block coefficients at the 5% level in nine specifications. WHZ contrasts are positive whenever the latest block contains more than a few thousand children, with joint significance at the 10% level in nine specifications. Later evening thresholds produce less precise estimates as sample sizes fall and standard errors increase. Overall, the block results support a weight pattern concentrated late in the fieldwork day rather than an artefact of linearity, while highlighting the compositional and precision limits of the latest comparisons.

6.4 Inference

6.4.1 Clustering of the standard errors

Standard errors are clustered at the DHS cluster level throughout the analysis as the cluster is the primary sampling unit. Since the treatment varies over time rather than space, however, households interviewed at the same moment might share unobserved conditions, making clock time a potential alternative clustering dimension. Appendix Table 19 reports the treatment coefficients under several clustering schemes. The point estimates are identical by construction, and the standard errors change very little. Clustering by the exact interview start minute increases the standard error for weight and WAZ, and two-way clustering on the DHS cluster and the start minute gives standard errors values that are between the two. All three weight-based coefficients remain significant at the one percent level, and height and HAZ remain statistically not significant, under every clustering scheme.

6.4.2 Leave-one-country-out

A pooled estimate could also be driven by a single country. Following the leave-one-country-out approach used in the literature, I re-estimate the main specification after excluding each of the 29 countries turn by turn. Figure 7 in the appendix plots the resulting estimates and confidence intervals against the full-sample estimate for each outcome. The weight-based coefficients remain robust across country-exclusion specifications. None changes sign, and all confidence intervals overlap the full-sample estimates, suggesting that the results are not driven by any single country. The estimates for height and HAZ remain statistically indistinguishable from zero across all specifications. Overall, this pattern suggests that the timing effect is robust for weight-based outcomes, with no systematic timing effect on height-based outcomes.

6.4.3 Leave-one-year-out

A similar concern relates to calendar time. Since the surveys were conducted over a decade, from 2015 to 2025, the pooled results could be influenced by an unusual year. To assess this possibility, appendix

Figure 8 reports a leave-one-year-out analysis, excluding one survey year at a time. The estimated effects on weight, WAZ, and WHZ remain positive across all specifications, indicating that the results are not driven by any single year. By contrast, the estimates for height and HAZ remain statistically insignificant and close to zero, with some variation in their signs across specifications. Overall, these findings suggest that the weight-related results are robust to the exclusion of individual survey years.

6.5 Reweighting the regression

The preceding robustness checks adjust for covariates by including them on the right-hand side of the regression and control for the survey weight. Entropy balancing provides an alternative approach as it constructs weights that make the covariate means of children measured earlier and later in the day similar. The treatment gradient is then estimated on this reweighted sample without relying on a regression model to perform the adjustment. Appendix Table 20 shows that the reweighting procedure achieves the intended balance. Standardized differences, which reach as high as 0.074 before reweighting, are reduced to zero after the weights are applied. Appendix Table 21 reports the corresponding estimates for each outcome. Under entropy balancing, the estimated gradient is consistent with the main estimates and none of the coefficients change by more than one-tenth of its own standard error. The corresponding estimates for height and HAZ are statistically not significant. The close correspondence between the two approaches indicates that the estimated weight gradient is not generated by the functional form imposed by the regression controls, and that the null results for height and HAZ are not produced by regression adjustment either.

6.6 Extending the sample with imputed timing

The main analysis is restricted to surveys conducted from 2015 onward, as earlier DHS rounds did not record the start time of the household questionnaire. These earlier surveys did, however, record the start time of the women's questionnaire, which makes it possible to approximate the missing household start time by predicting the gap between the two modules with a random forest trained on the 229,222 records of the 2015 and later surveys in which both timestamps are observed. The procedure adds 292,805 children from 80 surveys conducted before 2015. Appendix Table 22 reports the quality of the model and Appendix Figure 9 the variables it relies on. The model is stable but imprecise: its ten-fold cross-validated root mean squared error, 109 minutes, is the same as its hold-out error, so it generalizes well, but its mean absolute error of about 75 minutes is large relative to the less-than-four-hour interquartile range of observed start times.

Panel A of Appendix Table 23 reports the pooled estimates, with an indicator for the source of the timing measure. Among the 403,109 children in the estimation sample, recorded weight increases by 0.006 kilograms per hour, WHZ by 0.004 and WAZ by 0.003, all significant at the one percent level, and the probability of being classified underweight falls by 0.071 points per hour; the wasting gradient is zero in this sample. The height-based estimates are all statistically not significant. Panel B assesses the distortion that the imputation introduces. On the 2015 and later children for whom both timestamps are observed, replacing the actual household start time with its out-of-fold predicted value does not attenuate the estimates, as classical measurement error in a regressor would, but inflates them: the ratio of the imputed to the true coefficient is 1.65 for weight, 1.58 for WAZ, 1.29 for WHZ, 1.24 for wasting and 1.14 for underweight. The imputed estimates therefore cannot be read at face value, which is why the imputed sample is treated as a robustness exercise only and is not used for the main estimates.

6.7 An independent survey program

A final robustness test examines the same question in an independent data source. The MICS are implemented by UNICEF with separate field teams, training procedures and equipment. The replication sample comprises 69,824 children across 11 countries and 6,241 clusters; 59.3% are aged 24–59 months and 40.7% are under two (Appendix Table 24). Chad, Malawi and Madagascar account for approximately half the observations. Four countries (the Central African Republic, Chad, Eswatini and Togo) are absent from the DHS analytical sample. In the seven overlapping countries, survey years, questionnaires and field teams differ. Interview timing is nevertheless comparable: country-level mean start times range from 10:49 to 13:18, with most around 11:30, compared with 11:58 in the DHS.

The MICS estimates reproduce the main DHS patterns (Appendix Table 25). In the full sample, each additional hour is associated with increases of 0.007 kg in recorded weight and 0.005 in both WHZ and WAZ, all significant at the 5% level, and a 0.148-percentage-point decline in underweight. The wasting estimate is negative but insignificant. As in the DHS, these associations are concentrated among children aged 24–59 months: hourly increases reach 0.014 kg in weight, 0.010 in WHZ and 0.009 in WAZ, all significant at the 1% level, while underweight declines by 0.256 percentage points. None of the eight outcomes varies significantly with interview time among children under two, and height-based outcomes show no association in either age group.

The replication thus confirms both the weight and the height asymmetry and the concentration of the gradient among older children. Its appearance in partly distinct populations surveyed under separate operational arrangements makes an explanation specific to DHS practices less plausible and supports a broader interpretation of time-dependent anthropometric measurement.

7. Implications

The gradient documented above is a property of the measurement that enters two bodies of work: impact evaluations that use weight-based z-scores as outcomes, and the national prevalence statistics that surveys are designed to produce. This section works out what the gradient implies for each. The exercise is arithmetic and applies the estimated per-hour gradients of Table 4 to the hours over which measurement varies in practice and asks how the resulting shift compares with the effects the literature reports and with the prevalence rates the surveys publish.Table 4

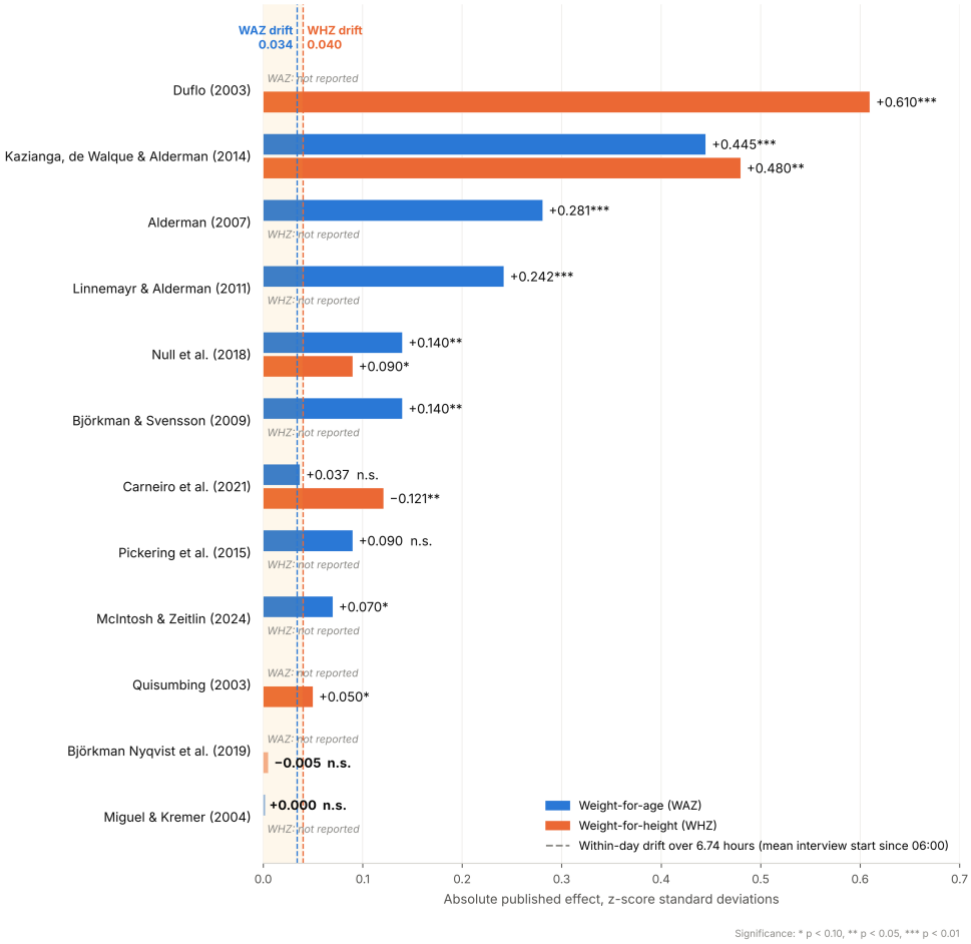
7.1 Implications for research on child nutrition

Across DHS surveys conducted from phase-7 onward, the mean interview time is 6.74 hours after the 06:00 benchmark, with a median of 6.33 hours. Applying the estimates in Table 4 implies that, at the mean interview time, a child would have a recorded WAZ approximately 0.03 standard deviations higher ($0.005 \times 6.74 = 0.034$) and a recorded WHZ approximately 0.04 standard deviations higher ($0.006 \times 6.74 = 0.040$) than at 06:00. Figure 4 compares this estimated time-of-day drift with effects reported in evaluations of child nutrition programs in SSA. Each row represents one study, and the two bars show the absolute magnitudes of its estimated effects on WAZ and WHZ. The figure show that direct food and nutrition program including take-home rations in Burkina Faso (Kazianga et al., 2014), community growth promotion in Uganda (Alderman, 2007), the Senegal nutrition enhancement project (Linnemayr & Alderman, 2011), the nutrition arm of the WASH benefits trial in Kenya (Null et al., 2018a) together with the South African old-age pension program (Duflo, 2003) reports an effects on weight-based indicators of 0.09 to 0.61 standard deviations, approximately two to fifteen times the corresponding drift. The remaining interventions comprise cash transfers in Nigeria and Rwanda

(Carneiro et al., 2021; McIntosh & Zeitlin, 2024); water, sanitation, and hygiene programs in Mali (Null et al., 2018b); community health promotion and deworming in Uganda and Kenya (Miguel & Kremer, 2004; Nyqvist et al., 2019); and food-for-work in Ethiopia (Quisumbing, 2003). Their estimated effects are approximately one to three times the size of the corresponding drift. Overall, two of the fifteen coefficients shown in the figure are smaller than the estimated shift between measurement at the 06:00 benchmark and at the mean interview time, indicating that the measurement-time drift can be larger than the estimated effects find in the literature.

The implications for research depend on whether measurement times differ between treatment and control groups. If the research design, fieldwork procedures or the treatment itself make treated households more or less likely to be surveyed at specific times of day, the resulting differences in measurement time can be mistaken for treatment effects. Because weight-based indicators change over the course of the day, treated children measured later than control children may appear to have better nutritional outcomes, even if the intervention has no effect. Conversely, if treated children are measured earlier, the timing difference may attenuate a genuine positive treatment effect making it appear smaller than it actually is. A difference of two to three hours in mean measurement time between treatment and control groups can generate an artefactual difference of 0.01 to 0.02 standard deviations in WHZ or WAZ. Researchers should therefore account for measurement time when estimating program effects to avoid attributing changes caused by the timing of measurement to the intervention itself.

Figure 4 : The within-day drift at the mean interview hour against programme evaluations of child nutrition



***Notes**: Each row is a published evaluation of a programme in one of the 29 countries of the analytical sample; blue bars are the effect on WAZ and orange bars the effect on WHZ, in z-score standard deviations and in absolute value, with the sign shown in the label; "not reported" marks an index*

*the study does not estimate. The dashed lines are the within-day drift implied by Table 4 (0.005 standard deviations per hour for WAZ and 0.006 for WHZ) over 6.74 hours, giving 0.034 and 0.040; 6.74 hours is the mean distance of a household interview from the 06:00 benchmark across the surveys fielded since 2015 (median 6.33 hours), so the lines mark the drift between the benchmark and the typical interview. Lighter bars are smaller than the drift for the same index. Where a study reports its significant effect for a subgroup, wave or arm, that estimate is shown and the pooled estimate is given here. Significance: * p < 0.10, ** p < 0.05, *** p < 0.01, as reported by each study or computed from its standard errors.*

7.2 Implications for national estimates of child undernutrition

The findings also have implications for national nutrition statistics. Children are classified as underweight or wasted when their WAZ or WHZ falls below −2. Because the recorded scores rise over the day, a child measured in the afternoon or evening can sit above the threshold when the same child measured in the morning would have sat below it. To quantify how much of the undernutrition that surveys would record at a morning benchmark is left out of the published prevalence, I move each child's weight-based z-scores to their 06:00 value by subtracting the hourly gradients of Table 4, 0.005 for WAZ and 0.006 for WHZ, multiplied by the hours elapsed since 06:00, apply the same thresholds, and count the children whose classification changes.[22] The exercise uses every measured child under five with a valid z-score in the 41 surveys that record the household start time, 263,732 children, so the first-visit and same-day restrictions of the estimation sample are not imposed, and it uses only DHS data: the numerator is the survey-weighted number of children whose classification flips and the denominator the survey-weighted number of children under five in the same surveys, with rounds of the same country averaged by their sample size.

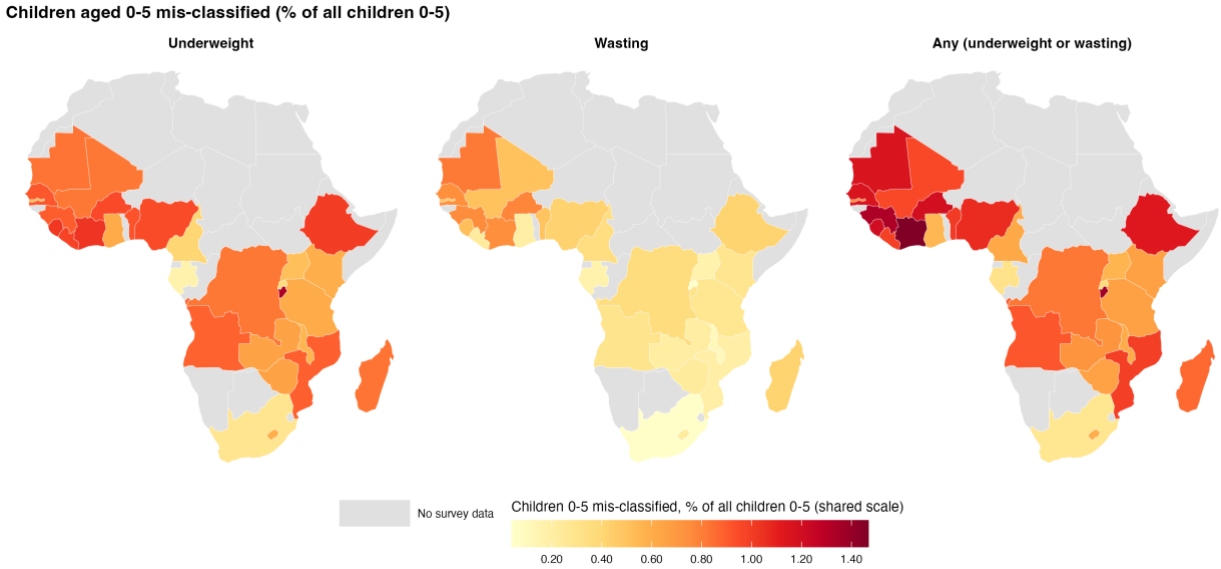
Across the 29 countries, moving the measurement to 06:00 increases the prevalence of underweight among children under five by 0.8 percentage points and that of wasting by 0.4 points, which is 4.6 and 5.9 percent of the respective prevalences at the benchmark; 0.9 percent of all children under five, one in twenty-one of those who would be classified undernourished in the morning, are recorded as adequately nourished because of the hour at which they were weighed. Applied to the under-five population of the 27 countries for which a denominator is available, 120.9 million children, these shares correspond to approximately 932,000 underweight children and 436,000 wasted children, 1.06 million children on either count, who are undernourished by the survey's own standard but recorded as adequately nourished. The same calculation at the −3 threshold reclassifies a further 338,000 children, 285,000 from severe to moderate underweight and 107,000 from severe to moderate wasting, whose severity rather than status is understated.[23] Figure 5 maps, for each country, the share of all children under five who are mis-classified, for underweight, for wasting and for either condition. The share is highest in Côte d'Ivoire (1.5 percent), Burundi (1.4), Guinea (1.3), Sierra Leone (1.2) and Burkina Faso (1.2), and lowest in South Africa (0.3), Gabon (0.3) and Rwanda (0.4). The patterns differ between the two indicators: the underweight share ranges from 0.2 points in Gabon to 1.3 in Burundi, and the wasting share from 0.05 in South Africa to 0.8 in Mauritania. The count of missed children follows population size and is concentrated in the most populous countries, with approximately 316,000 underweight children in Nigeria, 65,000 in Tanzania, 49,000 in Mozambique, 47,000 in Côte d'Ivoire, 45,000 in Angola and 42,000 in Kenya; for wasting the largest counts are again in Nigeria, with 143,000 children, followed by Côte d'Ivoire, Tanzania, Burkina Faso and Kenya. Appendix Table 26 gives the per-country values with their confidence intervals.

[22] For measurements taken after 06:00, this adjustment lowers z-scores and therefore either increases estimated undernutrition prevalence or leaves it unchanged. The exercise covers all children under five; the gradient applied is that of the full analytical sample (Panel A of Table 4).

[23] These estimates represent a lower bound for Sub-Saharan Africa because the analysis covers only 29 of the region's 48 countries, excludes some of the most populous countries, and does not include children under two years of age.

Appendix Figure 10 shows the subnational distribution of the understatement of underweight or wasting prevalence within each country, by first-level administrative unit, holding the national gradient fixed so that what varies across units is the local distribution of interview start times and the density of children close to the cut-off. The bias varies within countries, reaching two to four percentage points in some regions of Burkina Faso, Guinea, Mozambique, Tanzania, Côte d'Ivoire and Nigeria, and higher still in a few sparsely sampled units, where interviews take place later in the day or where many children sit close to the threshold. Overall, national statistics understate weight-based undernutrition, and the understatement is larger in the countries and survey rounds in which interviews are conducted later in the day. Stunting, the headline global indicator, is not affected: the HAZ gradient is 0.002 per hour, which translates into a change in stunting prevalence of 0.4 points at the benchmark, indistinguishable from zero.

Figure 5 : Children under five misclassified as adequately nourished by country in SSA



***Notes**: Share of all children under five in each country who are underweight (left), wasted (centre) or either (right) once weight-based z-scores are re-anchored to a 06:00 measurement, but are recorded as adequately nourished in the survey, in percent of the population under five. Moderate and severe cases are combined, that is children whose re-anchored z-score falls below −2 standard deviation. The pooled percentage-point bias for each country is estimated among all children aged 0–59 months with the DHS child weights and applied to the population under five. Countries in grey have no survey in the study sample. Each panel has its own colour scale. Per-country values and their confidence intervals are in Appendix Table 26.*

8. Conclusion

[Conclusion to be written.]

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Appendix

Table 5 : Construction of the study sample

Selection step	Children	Dropped
Pooled IPUMS-DHS extract plus the eight raw DHS 2023–25 surveys, Sub-Saharan Africa, all years	1,523,149	—
(i) Survey records a household interview start time (41 surveys, Phase 7 onward)	475,106	−1,048,043
(ii) Child measured: height and weight taken	267,406	−207,700
(iii) Valid start time, z-scores and measurement position (Ethiopia 2019 included, dates converted)	263,665	−3,741
Restriction: household completed on the first visit	223,594	−40,071
Restriction: measured on the day of the household interview	115,175	−108,419
Restriction: interview start time between 06:00 and 22:59	114,996	−179

***Notes**: The final row is the study sample. The first three rows are constants of the raw extract and of the recovery of the household start time from the raw DHS household files; the remaining rows are computed from the canonical build. Ethiopia enters through its 2019 survey, whose Ethiopian-calendar dates are converted; its 2016 survey drops at the same-day step because the extract carries no measurement date for it. Individual outcomes use their own complete cases; height and weight are available for all 114,996 children, and the z-scores lose a few hundred records to the plausibility screen.*

Table 6 : Study sample by country and survey year

Country	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	Total
Angola	3,643								2,100			5,743
Benin			4,943									4,943
Burkina Faso							2,114					2,114
Burundi		2,464										2,464
Cameroon				2,096								2,096
Congo, Dem. Rep.									3,306			3,306
Côte d'Ivoire							2,312					2,312
Ethiopia					3,235					1,732		4,967
Gabon					2,623							2,623
Gambia					491							491
Ghana								1,761				1,761
Guinea				1,672								1,672
Kenya								9,802				9,802
Lesotho									464			464
Liberia					606							606
Madagascar							4,495					4,495
Malawi	2,734									2,747		5,481
Mali				2,017					4,679			6,696
Mauritania						3,917						3,917

Country	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	Total
Mozambique								1,553				1,553
Nigeria				5,192						2,726		7,918
Rwanda					1,677						3,756	5,433
Senegal			2,750	2,586	2,150				1,149			8,635
Sierra Leone					1,609							1,609
South Africa			345									345
Tanzania	7,761							2,443				10,204
Uganda		2,542										2,542
Zambia				4,958						2,516		7,474
Zimbabwe	3,330											3,330
Total	17,468	5,006	8,038	18,521	12,391	3,917	8,921	15,559	11,698	9,721	3,756	114,996

***Notes**: Number of children in the study sample by country and survey year. Blank cells indicate no survey retained for that country-year.*

Table 7 : Study sample compared with all other DHS children

Characteristic	Study sample	All other DHS	Difference	Norm. diff.
Child age (months)	28.087	27.804	0.283***	0.016
Female	0.500	0.496	0.004*	0.007
Aged 24–59 months	0.532	0.555	−0.023***	−0.045
Birth order	3.368	3.663	−0.295***	−0.126
Twin (multiple)	0.031	0.029	0.001*	0.007
Mother's age (years)	29.286	29.042	0.243***	0.035
Parity	3.656	3.985	−0.329***	−0.138
Mother's education (years)	5.241	4.014	1.227***	0.270
Mother's education				
None	0.310	0.438	−0.128***	−0.267
Primary	0.369	0.330	0.039***	0.082
Secondary	0.271	0.202	0.069***	0.162
Higher	0.051	0.031	0.020***	0.102
Missing	0.000	0.000	−0.000***	−0.011
Father's education (years)	6.002	5.059	0.943***	0.185
Father's education				
None	0.240	0.345	−0.105***	−0.232
Primary	0.180	0.207	−0.026***	−0.066
Secondary	0.300	0.276	0.024***	0.052
Higher	0.074	0.065	0.008***	0.032
Missing	0.206	0.107	0.099***	0.275
Wealth quintile				
Poorest	0.231	0.225	0.007***	0.016
Second	0.213	0.212	0.002	0.004
Third	0.197	0.204	−0.008***	−0.019
Fourth	0.191	0.192	−0.002	−0.004
Richest	0.168	0.167	0.001	0.003
Urban	0.324	0.290	0.035***	0.075
Household size	7.079	7.579	−0.500***	−0.106
No. children < 5	2.076	2.228	−0.152***	−0.111
Head's age	40.792	41.239	−0.446***	−0.033
Female-headed household	0.239	0.167	0.073***	0.181
Height-for-age z-score (HAZ)	−1.250	−1.450	0.200***	0.130
Weight-for-age z-score (WAZ)	−0.805	−0.941	0.136***	0.114
Weight-for-height z-score (WHZ)	−0.135	−0.156	0.022***	0.017
Height (cm)	83.238	81.989	1.249***	0.089
Weight (kg)	11.202	10.939	0.264***	0.075
Children	114,996	740,855		

Characteristic	Study sample	All other DHS	Difference	Norm. diff.
Surveys	40	157		
Countries	29	47		

***Notes**: Survey-weighted means. The study sample is the estimation sample; the comparison column is every other child with valid anthropometry in the extract (157 surveys in 47 Sub-Saharan countries, 1991–2025), comprising the pre-2015 rounds and the children removed by the restrictions. The two columns are disjoint, and the comparison group is not filtered on first visit or measurement day. Standard errors behind the stars use Kish effective sample sizes. The final column is the Imbens–Rubin normalized difference; values above 0.25 in absolute terms indicate a substantive difference. Parental education is shown both in years and as categories. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.*

Table 8 : Heterogeneity in the timing gradient

	Weight-based outcomes					Height-based outcomes		
Moderator / term	**(1) Wasted (pp)**	**(2) WHZ**	**(3) Underweight (pp)**	**(4) WAZ**	**(5) Weight (kg)**	**(6) Stunted (pp)**	**(7) HAZ**	**(8) Height (cm)**
Panel A. Child sex								
Hours since 06:00 (base group)	−0.079 (0.051)	0.006** (0.003)	−0.121 (0.080)	0.003 (0.003)	0.007* (0.004)	0.079 (0.102)	−0.002 (0.003)	−0.005 (0.011)
Hours since 06:00 × Female	0.006 (0.062) [1.000]	−0.001 (0.003) [1.000]	0.024 (0.095) [1.000]	0.003 (0.003) [1.000]	0.006 (0.005) [1.000]	−0.170 (0.127) [1.000]	0.008** (0.004) [0.857]	0.029** (0.014) [0.786]
Marginal effect at female	−0.073 (0.047)	0.005* (0.003)	−0.097 (0.077)	0.006** (0.003)	0.013*** (0.004)	−0.091 (0.096)	0.006* (0.003)	0.024** (0.010)
R^2	0.203	0.293	0.243	0.332	0.857	0.293	0.350	0.927
Observations	114,052	114,052	114,489	114,489	114,550	114,550	114,550	114,550
Panel B. Twin (multiple birth)								
Hours since 06:00 (base group)	−0.075** (0.038)	0.006*** (0.002)	−0.114* (0.062)	0.005** (0.002)	0.011*** (0.003)	0.014 (0.076)	0.002 (0.002)	0.009 (0.008)
Hours since 06:00 × Twin	−0.039 (0.263) [1.000]	−0.013 (0.012) [1.000]	0.177 (0.429) [1.000]	−0.004 (0.012) [1.000]	−0.022 (0.017) [1.000]	−0.727 (0.490) [1.000]	0.008 (0.015) [1.000]	0.027 (0.053) [1.000]
Marginal effect at twin	−0.114 (0.264)	−0.007 (0.012)	0.064 (0.430)	0.000 (0.012)	−0.012 (0.017)	−0.713 (0.489)	0.009 (0.015)	0.036 (0.053)
R^2	0.203	0.293	0.243	0.332	0.857	0.293	0.350	0.927
Observations	114,052	114,052	114,489	114,489	114,550	114,550	114,550	114,550
Panel C. Birth order								
Hours since 06:00 (base group)	−0.097** (0.042)	0.007*** (0.002)	−0.127* (0.068)	0.006*** (0.002)	0.011*** (0.003)	−0.015 (0.084)	0.002 (0.003)	0.008 (0.009)
Hours since 06:00 × First-born	0.090 (0.076) [1.000]	−0.007* (0.004) [0.859]	0.077 (0.113) [1.000]	−0.005 (0.004) [1.000]	−0.003 (0.005) [1.000]	0.040 (0.146) [1.000]	−0.001 (0.004) [1.000]	0.008 (0.015) [1.000]
Marginal effect at first-born	−0.007 (0.070)	0.000 (0.004)	−0.050 (0.107)	0.001 (0.003)	0.008 (0.005)	0.026 (0.133)	0.001 (0.004)	0.016 (0.014)
R^2	0.203	0.293	0.243	0.332	0.857	0.293	0.350	0.927
Observations	114,052	114,052	114,489	114,489	114,550	114,550	114,550	114,550
Panel D. Age in months								
Hours since 06:00 (base group)	−0.010 (0.065)	0.003 (0.004)	−0.135 (0.092)	0.003 (0.003)	0.001 (0.004)	0.070 (0.118)	0.002 (0.004)	−0.007 (0.013)
Hours since 06:00 × Age (months)	−0.002 (0.002) [1.000]	0.000 (0.000) [1.000]	0.001 (0.003) [1.000]	0.000 (0.000) [1.000]	0.000** (0.000) [0.767]	−0.003 (0.003) [1.000]	0.000 (0.000) [1.000]	0.001 (0.000) [1.000]
Marginal effect at 13 months (p25)	−0.040 (0.048)	0.004* (0.003)	−0.123* (0.071)	0.004 (0.003)	0.005 (0.003)	0.035 (0.090)	0.002 (0.003)	0.000 (0.009)
Marginal effect at 43 months (p75)	−0.110** (0.046)	0.007*** (0.002)	−0.095 (0.076)	0.006** (0.002)	0.015*** (0.004)	−0.045 (0.092)	0.002 (0.003)	0.019* (0.011)
R^2	0.203	0.293	0.243	0.332	0.857	0.293	0.350	0.927
Observations	114,052	114,052	114,489	114,489	114,550	114,550	114,550	114,550
Panel E. Age group: 24–59 months versus 0–23 months								
Hours since 06:00 (base group)	−0.018 (0.055)	0.005 (0.003)	−0.088 (0.081)	0.003 (0.003)	0.003 (0.003)	0.124 (0.100)	0.000 (0.003)	−0.010 (0.010)
Hours since 06:00 × Aged 24–59 months	−0.104 (0.064) [1.000]	0.002 (0.003) [1.000]	−0.041 (0.094) [1.000]	0.003 (0.003) [1.000]	0.013*** (0.005) [0.544]	−0.232* (0.122) [0.859]	0.003 (0.004) [1.000]	0.030** (0.013) [0.767]

	Weight-based outcomes					Height-based outcomes		
Moderator / term	**(1) Wasted (pp)**	**(2) WHZ**	**(3) Underweight (pp)**	**(4) WAZ**	**(5) Weight (kg)**	**(6) Stunted (pp)**	**(7) HAZ**	**(8) Height (cm)**
Marginal effect at 24–59 months	−0.122*** (0.046)	0.007*** (0.002)	−0.129* (0.076)	0.006*** (0.002)	0.015*** (0.004)	−0.108 (0.094)	0.003 (0.003)	0.021* (0.011)
R^2	0.204	0.299	0.243	0.334	0.858	0.294	0.351	0.929
Observations	114,052	114,052	114,489	114,489	114,550	114,550	114,550	114,550
Panel F. Household wealth								
Hours since 06:00 (base group)	−0.121** (0.054)	0.006** (0.003)	−0.043 (0.081)	0.004 (0.003)	0.012** (0.005)	0.128 (0.098)	0.000 (0.003)	0.005 (0.011)
Hours since 06:00 × Poorest two quintiles	0.098 (0.074) [1.000]	−0.001 (0.004) [1.000]	−0.144 (0.116) [1.000]	0.002 (0.004) [1.000]	−0.003 (0.006) [1.000]	−0.291** (0.145) [0.857]	0.004 (0.004) [1.000]	0.010 (0.016) [1.000]
Marginal effect at poorest two quintiles	−0.023 (0.052)	0.005* (0.003)	−0.187** (0.090)	0.006** (0.003)	0.009** (0.004)	−0.163 (0.111)	0.004 (0.003)	0.015 (0.011)
R^2	0.203	0.293	0.243	0.332	0.857	0.293	0.350	0.927
Observations	114,052	114,052	114,489	114,489	114,550	114,550	114,550	114,550
Panel G. Residence								
Hours since 06:00 (base group)	−0.026 (0.043)	0.004* (0.002)	−0.095 (0.076)	0.003 (0.002)	0.006* (0.004)	−0.024 (0.093)	−0.000 (0.003)	−0.001 (0.009)
Hours since 06:00 × Urban	−0.155* (0.088) [0.884]	0.005 (0.005) [1.000]	−0.045 (0.133) [1.000]	0.005 (0.005) [1.000]	0.012* (0.007) [0.884]	0.057 (0.158) [1.000]	0.006 (0.005) [1.000]	0.033* (0.018) [0.859]
Marginal effect at urban	−0.181** (0.077)	0.009** (0.004)	−0.140 (0.110)	0.008** (0.004)	0.019*** (0.006)	0.033 (0.128)	0.006 (0.004)	0.032** (0.016)
R^2	0.203	0.293	0.243	0.332	0.857	0.293	0.350	0.927
Observations	114,052	114,052	114,489	114,489	114,550	114,550	114,550	114,550

***Notes**: All children aged 0–59 months. Each panel adds to the main specification the interaction of hours since 06:00 with one moderator; the first row is the gradient in the base group, the second the interaction, and the marginal-effect rows the gradient implied for the other group, with its own standard error. Age in months is evaluated at the 25th and 75th percentiles of the sample; the age-group indicator equals one for children aged 0–59 months, who are measured standing. Values in square brackets are Anderson (2008) sharpened q-values computed across all interaction terms of the heterogeneity analysis as one family. The full control set is the child's sex, age in months and its square, birth order and twin status; the mother's age, parity and level of education; the father's education; the household's wealth quintile, urban residence, size and number of children under five; the age and sex of the household head; and the length of the household interview module. Every regression includes DHS cluster and day-of-week fixed effects, uses the DHS child weights, and clusters standard errors at the DHS cluster. Wasted, underweight and stunted are linear probability models for the indicator that the z-score falls below −2, with the coefficient scaled to percentage points. *** p < 0.01, ** p < 0.05, * p < 0.1.*

Table 9 : Interview-day temperature and the timing gradient

	Weight-based outcomes					Height-based outcomes		
Term	**(1) Wasted (pp)**	**(2) WHZ**	**(3) Underweight (pp)**	**(4) WAZ**	**(5) Weight (kg)**	**(6) Stunted (pp)**	**(7) HAZ**	**(8) Height (cm)**
Panel A. Hours since 06:00 × interview-day wet-bulb temperature								
Hours since 06:00	−0.033 (0.159)	0.010 (0.009)	0.218 (0.277)	0.010 (0.009)	0.026* (0.014)	0.043 (0.356)	0.005 (0.010)	0.016 (0.036)
Hours since 06:00 × Wet-bulb (°C)	−0.002 (0.009) [1.000]	−0.000 (0.000) [1.000]	−0.018 (0.015) [1.000]	−0.000 (0.000) [1.000]	−0.001 (0.001) [1.000]	−0.002 (0.018) [1.000]	−0.000 (0.001) [1.000]	−0.001 (0.002) [1.000]
Marginal effect at 14.9 °C (p25)	−0.063 (0.047)	0.007** (0.003)	−0.045 (0.081)	0.006** (0.003)	0.013*** (0.004)	0.018 (0.109)	0.001 (0.003)	0.007 (0.011)
Marginal effect at 22.4 °C (p75)	−0.079 (0.054)	0.006** (0.003)	−0.178** (0.088)	0.004 (0.003)	0.007* (0.004)	0.005 (0.098)	−0.000 (0.003)	0.003 (0.011)
Observations	107,919	107,919	108,315	108,315	108,369	108,369	108,369	108,369
Panel B. Hot versus cool interview days								
Hours since 06:00 (cool day)	−0.068 (0.047)	0.009*** (0.003)	−0.129 (0.083)	0.006** (0.003)	0.013*** (0.004)	−0.009 (0.105)	0.000 (0.003)	0.003 (0.011)
Hours since 06:00 × Hot day	−0.008 (0.081) [1.000]	−0.006 (0.004) [1.000]	0.046 (0.130) [1.000]	−0.004 (0.004) [1.000]	−0.007 (0.006) [1.000]	0.049 (0.158) [1.000]	0.000 (0.005) [1.000]	0.005 (0.017) [1.000]
Marginal effect, hot day	−0.076 (0.066)	0.003 (0.003)	−0.083 (0.100)	0.002 (0.003)	0.006 (0.005)	0.039 (0.118)	0.001 (0.004)	0.008 (0.013)
Observations	107,919	107,919	108,315	108,315	108,369	108,369	108,369	108,369

***Notes**: All children aged 0–59 months, full-control specification. Panel A interacts hours since 06:00 with the interview-day wet-bulb temperature at the cluster (ERA5); the cluster fixed effects absorb the local climatological level, so the interaction is identified from day-to-day variation within a cluster. Panel B interacts the gradient with an indicator for a hot interview day (dry-bulb temperature above the cluster's 2000–2020 climatological mean). Values in square brackets are Anderson (2008) sharpened q-values computed across all interaction terms of the heterogeneity analysis as one family. The full control set is the child's sex, age in months and its square, birth order and twin status; the mother's age, parity and level of education; the father's education; the household's wealth quintile, urban residence, size and number of children under five; the age and sex of the household head; and the length of the household interview module. Every regression includes DHS cluster and day-of-week fixed effects, uses the DHS child weights, and clusters standard errors at the DHS cluster. Wasted, underweight and stunted are linear probability models for the indicator that the z-score falls below −2, with the coefficient scaled to percentage points. *** p < 0.01, ** p < 0.05, * p < 0.1.*

Table 10 : Mechanism to digit heaping in the recorded measurement

	Weight-based outcomes					Height-based outcomes		
Term	**(1) Wasted (pp)**	**(2) WHZ**	**(3) Underweight (pp)**	**(4) WAZ**	**(5) Weight (kg)**	**(6) Stunted (pp)**	**(7) HAZ**	**(8) Height (cm)**
Hours since 06:00	−0.076** (0.038) [0.064]	0.006*** (0.002) [0.018]	−0.109* (0.063) [0.089]	0.005** (0.002) [0.042]	0.010*** (0.003) [0.008]	−0.005 (0.076) [0.468]	0.002 (0.002) [0.218]	0.010 (0.008) [0.150]
Measurement ends in .0 (heaped)	0.541** (0.215)	−0.023** (0.011)	0.083 (0.331)	−0.000 (0.010)	−0.014 (0.015)	−0.409 (0.413)	0.019 (0.012)	0.030 (0.043)
Mean of outcome	4.961	−0.163	13.289	−0.826	11.168	27.754	−1.251	83.216
R^2	0.203	0.293	0.243	0.332	0.857	0.293	0.350	0.927
Observations	114,052	114,052	114,489	114,489	114,550	114,550	114,550	114,550

***Notes**: All children aged 0–59 months. The main specification with an added indicator for a recorded weight or height ending in .0, the signature of rounding. The full control set is the child's sex, age in months and its square, birth order and twin status; the mother's age, parity and level of education; the father's education; the household's wealth quintile, urban residence, size and number of children under five; the age and sex of the household head; and the length of the household interview module. Every regression includes DHS cluster and day-of-week fixed effects, uses the DHS child weights, and clusters standard errors at the DHS cluster. Wasted, underweight and stunted are linear probability models for the indicator that the z-score falls below −2, with the coefficient scaled to percentage points. Values in square brackets are Anderson (2008) sharpened false-discovery-rate q-values, computed within the family of the eight outcomes of the panel. *** p < 0.01, ** p < 0.05, * p < 0.1.*

Table 11 : Does measurement noise rise over the day?

Outcome	Coefficient on hours since 06:00	on Afternoon	on Evening	Mean of outcome	Observations
Panel A. Squared residual of the main specification					
WAZ	−0.001 (0.003)	0.010 (0.013)	−0.050** (0.025)	0.815	114,489
WHZ	−0.004 (0.003)	0.013 (0.016)	−0.085*** (0.029)	0.913	114,052
HAZ	0.005 (0.004)	0.011 (0.022)	0.032 (0.042)	1.193	114,550
Panel B. Probability of a data-quality failure (coefficients × 100)					
Flagged z-score	0.006 (0.013)	−0.002 (0.087)	0.041 (0.112)	0.011	179,474
Implausible z-score	0.003 (0.005)	0.022 (0.034)	0.043 (0.046)	0.002	179,474
Missing height	0.034 (0.023)	0.003 (0.139)	0.149 (0.190)	0.027	179,474
Missing weight	0.035 (0.023)	0.037 (0.137)	0.150 (0.186)	0.026	179,474
Missing height or weight	0.036 (0.023)	0.004 (0.139)	0.162 (0.190)	0.027	179,474

***Notes**: Panel A regresses the squared residuals from the main full-control specification, children aged 0–59 months, on the treatment with cluster and day-of-week fixed effects and standard errors clustered at the DHS cluster; a positive coefficient would mean measurements become noisier later in the day. Panel B reports linear probability models of each data-quality failure on the treatment, with the full control set of Table 4, DHS cluster and day-of-week fixed effects, survey weights and standard errors clustered at the DHS cluster; coefficients and standard errors are multiplied by 100 and therefore read as percentage points. The universe of Panel B is the 193,822 children in the anthropometry module of the study surveys, measured or not, on a first visit with a start time in the valid window (179,474 complete on every control); it deliberately does not impose the measured or same-day restrictions, since a measured child has height and weight by construction. Flagged and implausible z-scores follow the DHS quality flags; the mean of outcome is the share of children with the failure. A positive coefficient would mean that later measurements fail quality checks more often, the enumerator-fatigue prediction. *** p < 0.01, ** p < 0.05, * p < 0.1.*

Table 12 : Robustness to fixed effects, weighting and controls

	Weight-based outcomes					Height-based outcomes		
Specification	**(1) Wasted (pp)**	**(2) WHZ**	**(3) Underweight (pp)**	**(4) WAZ**	**(5) Weight (kg)**	**(6) Stunted (pp)**	**(7) HAZ**	**(8) Height (cm)**
Panel A. Unweighted (survey weights dropped)								
Hours since 06:00	−0.025 (0.033) [0.425]	0.006*** (0.002) [0.002]	−0.100** (0.050) [0.059]	0.004*** (0.002) [0.011]	0.010*** (0.003) [0.001]	−0.026 (0.063) [0.514]	0.001 (0.002) [0.440]	0.008 (0.007) [0.272]
Mean of outcome	4.961	−0.163	13.289	−0.826	11.168	27.754	−1.251	83.216
R^2	0.205	0.297	0.242	0.325	0.857	0.290	0.341	0.926
Observations	114,052	114,052	114,489	114,489	114,550	114,550	114,550	114,550
Panel B. Age-in-months fixed effects								
Hours since 06:00	−0.075* (0.043) [0.088]	0.005** (0.002) [0.077]	−0.119* (0.066) [0.088]	0.005** (0.002) [0.077]	0.010*** (0.003) [0.038]	−0.011 (0.080) [0.357]	0.003 (0.003) [0.145]	0.009 (0.009) [0.149]
Mean of outcome	5.407	−0.191	14.183	−0.846	11.147	28.084	−1.252	83.223
R^2	0.201	0.298	0.240	0.338	0.858	0.298	0.354	0.931
Observations	90,738	90,738	91,150	91,150	91,211	91,211	91,211	91,211
Panel C. Month × country fixed effects added to the cluster fixed effects								

	Weight-based outcomes					Height-based outcomes		
Specification	**(1) Wasted (pp)**	**(2) WHZ**	**(3) Underweight (pp)**	**(4) WAZ**	**(5) Weight (kg)**	**(6) Stunted (pp)**	**(7) HAZ**	**(8) Height (cm)**
Hours since 06:00	−0.072* (0.038) [0.082]	0.006*** (0.002) [0.021]	−0.106* (0.063) [0.102]	0.005** (0.002) [0.049]	0.010*** (0.003) [0.010]	0.001 (0.076) [0.497]	0.002 (0.002) [0.234]	0.009 (0.008) [0.173]
Mean of outcome	4.961	−0.163	13.289	−0.826	11.168	27.754	−1.251	83.216
R^2	0.204	0.294	0.244	0.333	0.858	0.294	0.351	0.927
Observations	114,052	114,052	114,489	114,489	114,550	114,550	114,550	114,550
Panel D. Interview-day wet-bulb temperature controlled								
Hours since 06:00	−0.069* (0.038) [0.082]	0.006*** (0.002) [0.017]	−0.113* (0.063) [0.082]	0.005** (0.002) [0.036]	0.011*** (0.003) [0.006]	−0.005 (0.077) [0.554]	0.002 (0.002) [0.283]	0.009 (0.008) [0.178]
Mean of outcome	4.958	−0.161	13.370	−0.831	11.159	28.011	−1.262	83.173
R^2	0.203	0.292	0.242	0.328	0.857	0.292	0.346	0.927
Observations	110,779	110,779	111,192	111,192	111,252	111,252	111,252	111,252
Panel E. Women's questionnaire length controlled								
Hours since 06:00	−0.081** (0.039) [0.050]	0.006*** (0.002) [0.016]	−0.101 (0.063) [0.099]	0.005** (0.002) [0.032]	0.011*** (0.003) [0.004]	−0.001 (0.077) [0.328]	0.002 (0.002) [0.220]	0.012 (0.008) [0.121]
Mean of outcome	4.923	−0.162	13.279	−0.826	11.170	27.755	−1.252	83.227
R^2	0.203	0.294	0.243	0.333	0.857	0.295	0.350	0.927
Observations	110,485	110,485	110,903	110,903	110,962	110,962	110,962	110,962
Panel F. Total household interview duration controlled								
Hours since 06:00	−0.075* (0.038) [0.069]	0.006*** (0.002) [0.021]	−0.107* (0.063) [0.097]	0.005** (0.002) [0.041]	0.010*** (0.003) [0.008]	−0.008 (0.076) [0.429]	0.002 (0.002) [0.200]	0.010 (0.008) [0.163]
Mean of outcome	4.961	−0.163	13.290	−0.826	11.168	27.755	−1.251	83.216
R^2	0.203	0.293	0.243	0.332	0.857	0.293	0.350	0.927
Observations	114,025	114,025	114,462	114,462	114,523	114,523	114,523	114,523

***Notes**: All children aged 0–59 months. Each panel re-estimates the main specification with the stated change. The full control set is the child's sex, age in months and its square, birth order and twin status; the mother's age, parity and level of education; the father's education; the household's wealth quintile, urban residence, size and number of children under five; the age and sex of the household head; and the length of the household interview module. Every regression includes DHS cluster and day-of-week fixed effects, uses the DHS child weights, and clusters standard errors at the DHS cluster. Wasted, underweight and stunted are linear probability models for the indicator that the z-score falls below −2, with the coefficient scaled to percentage points. Values in square brackets are Anderson (2008) sharpened false-discovery-rate q-values, computed within the family of the eight outcomes of the panel. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.*

Table 13 : Robustness to the same-day measurement restriction

	Weight-based outcomes					Height-based outcomes		
Sample definition	**(1) Wasted (pp)**	**(2) WHZ**	**(3) Underweight (pp)**	**(4) WAZ**	**(5) Weight (kg)**	**(6) Stunted (pp)**	**(7) HAZ**	**(8) Height (cm)**
Panel A. Measured on the same or the next day								
Hours since 06:00	−0.043 (0.031) [0.093]	0.001 (0.002) [0.179]	−0.134*** (0.048) [0.033]	0.003** (0.002) [0.040]	0.006*** (0.002) [0.033]	−0.057 (0.059) [0.167]	0.004** (0.002) [0.033]	0.016** (0.007) [0.033]
Mean of outcome	5.767	−0.221	14.616	−0.866	11.133	28.079	−1.248	83.275
R^2	0.158	0.247	0.195	0.289	0.835	0.255	0.309	0.916
Observations	134,909	134,909	135,517	135,517	135,611	135,611	135,611	135,611
Panel B. Measured on any day, interview-to-measurement delay controlled								
Hours since 06:00	−0.031 (0.029) [0.234]	0.001 (0.001) [0.252]	−0.109** (0.044) [0.113]	0.002 (0.001) [0.138]	0.004** (0.002) [0.113]	−0.026 (0.054) [0.312]	0.003* (0.002) [0.128]	0.012** (0.006) [0.113]
Mean of outcome	6.160	−0.237	15.382	−0.888	11.116	28.701	−1.266	83.249
R^2	0.143	0.232	0.179	0.272	0.829	0.237	0.291	0.911
Observations	163,446	163,446	164,228	164,228	164,349	164,349	164,349	164,349
Panel C. Measured on any day, delay dummies and visit count added as controls								
Hours since 06:00	−0.086*** (0.030) [0.007]	0.002 (0.002) [0.035]	−0.189*** (0.045) [0.001]	0.004*** (0.001) [0.007]	0.006*** (0.002) [0.007]	−0.121** (0.055) [0.010]	0.005*** (0.002) [0.008]	0.018*** (0.006) [0.007]
Mean of outcome	6.160	−0.237	15.382	−0.888	11.116	28.701	−1.266	83.249
R^2	0.145	0.232	0.181	0.273	0.829	0.239	0.292	0.911
Observations	163,446	163,446	164,228	164,228	164,349	164,349	164,349	164,349

***Notes**: All children aged 0–59 months. The main sample keeps children measured on the day of the household interview; each panel relaxes that restriction as stated. The full control set is the child's sex, age in months and its square, birth order and twin status; the mother's age, parity and level of education; the father's education; the household's wealth quintile, urban residence, size and number of children under five; the age and sex of the household head; and the length of the household interview module. Every regression includes DHS cluster and day-of-week fixed effects, uses the DHS child weights, and clusters standard errors at the DHS cluster. Wasted, underweight and stunted are linear probability models for the indicator that the z-score falls below −2, with the coefficient scaled to percentage points. Values in square brackets are Anderson (2008) sharpened false-discovery-rate q-values, computed within the family of the eight outcomes of the panel. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.*

Table 14 : Robustness to the sample period and to the anchor of the treatment

	Weight-based outcomes					Height-based outcomes		
Specification	**(1) Wasted (pp)**	**(2) WHZ**	**(3) Underweight (pp)**	**(4) WAZ**	**(5) Weight (kg)**	**(6) Stunted (pp)**	**(7) HAZ**	**(8) Height (cm)**
Panel A. Pre-COVID surveys only (19 of 40)								
Hours since 06:00	−0.046 (0.052) [0.799]	0.003 (0.003) [0.799]	−0.106 (0.085) [0.799]	0.003 (0.003) [0.799]	0.009** (0.004) [0.514]	−0.052 (0.102) [1.000]	0.002 (0.003) [0.881]	0.011 (0.011) [0.799]
Mean of outcome	5.602	−0.164	15.695	−0.892	11.077	31.723	−1.360	82.807
R^2	0.181	0.258	0.219	0.309	0.840	0.274	0.329	0.912
Observations	56,068	56,068	56,427	56,427	56,482	56,482	56,482	56,482
Panel B. Excluding surveys with an anomalous measurement-position protocol (38 of 40)								
Hours since 06:00	−0.083** (0.040) [0.051]	0.006*** (0.002) [0.023]	−0.110* (0.064) [0.078]	0.005** (0.002) [0.038]	0.011*** (0.003) [0.007]	−0.018 (0.077) [0.303]	0.002 (0.002) [0.158]	0.012 (0.008) [0.132]
Mean of outcome	5.005	−0.154	13.452	−0.826	11.165	28.078	−1.260	83.167
R^2	0.207	0.298	0.246	0.337	0.856	0.297	0.352	0.926
Observations	107,645	107,645	108,081	108,081	108,142	108,142	108,142	108,142
Panel C. Treatment measured from 07:00, interviews starting at or after 07:00								
Hours since 07:00	−0.069* (0.039) [0.110]	0.005** (0.002) [0.048]	−0.102 (0.062) [0.113]	0.004** (0.002) [0.066]	0.010*** (0.003) [0.012]	0.003 (0.077) [0.394]	0.002 (0.002) [0.196]	0.010 (0.008) [0.189]
Mean of outcome	4.961	−0.162	13.264	−0.825	11.169	27.728	−1.250	83.218
R^2	0.203	0.293	0.244	0.332	0.857	0.294	0.351	0.927
Observations	113,253	113,253	113,685	113,685	113,746	113,746	113,746	113,746
Panel D. Treatment measured from 08:00, interviews starting at or after 08:00								
Hours since 08:00	−0.065 (0.041) [0.171]	0.005** (0.002) [0.101]	−0.083 (0.064) [0.231]	0.004* (0.002) [0.123]	0.010*** (0.003) [0.023]	0.018 (0.081) [0.428]	0.002 (0.002) [0.365]	0.010 (0.009) [0.231]
Mean of outcome	4.944	−0.161	13.212	−0.823	11.170	27.642	−1.248	83.220
R^2	0.208	0.298	0.249	0.336	0.858	0.297	0.354	0.928
Observations	109,744	109,744	110,161	110,161	110,222	110,222	110,222	110,222

***Notes**: All children aged 0–59 months. Panel A keeps the surveys whose fieldwork ended before 11 March 2020; Panel B drops the two surveys (Mali 2023, Ethiopia 2024) in which the measurement position does not follow the 24-month protocol; Panels C and D re-anchor the continuous treatment at 07:00 and 08:00 and drop interviews that started earlier. The full control set is the child's sex, age in months and its square, birth order and twin status; the mother's age, parity and level of education; the father's education; the household's wealth quintile, urban residence, size and number of children under five; the age and sex of the household head; and the length of the household interview module. Every regression includes DHS cluster and day-of-week fixed effects, uses the DHS child weights, and clusters standard errors at the DHS cluster. Wasted, underweight and stunted are linear probability models for the indicator that the z-score falls below −2, with the coefficient scaled to percentage points. Values in square brackets are Anderson (2008) sharpened false-discovery-rate q-values, computed within the family of the eight outcomes of the panel. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.*

Table 15 : Alternative definition of the treatment: time-of-day blocks relative to the morning

	Weight-based outcomes					Height-based outcomes		
Time-of-day block	**(1) Wasted (pp)**	**(2) WHZ**	**(3) Underweight (pp)**	**(4) WAZ**	**(5) Weight (kg)**	**(6) Stunted (pp)**	**(7) HAZ**	**(8) Height (cm)**
Afternoon (12:00–15:59)	−0.110 (0.201) [0.479]	0.014 (0.011) [0.198]	0.069 (0.331) [0.492]	0.005 (0.011) [0.479]	0.023 (0.015) [0.125]	0.466 (0.419) [0.240]	−0.006 (0.012) [0.479]	0.008 (0.043) [0.492]
Evening (16:00–22:59)	−0.951** (0.373) [0.019]	0.052*** (0.020) [0.019]	−2.111*** (0.598) [0.003]	0.071*** (0.020) [0.003]	0.124*** (0.033) [0.003]	−1.474** (0.721) [0.048]	0.067*** (0.024) [0.015]	0.229*** (0.084) [0.017]
Mean of outcome	4.961	−0.163	13.289	−0.826	11.168	27.754	−1.251	83.216
R^2	0.203	0.293	0.243	0.332	0.857	0.293	0.350	0.927
Observations	114,052	114,052	114,489	114,489	114,550	114,550	114,550	114,550

***Notes**: All children aged 0–59 months. Each cell is the coefficient on the indicator for an interview started in the afternoon (12:00–15:59) or in the evening (16:00–22:59), relative to one started in the morning (06:00–11:59), from a separate regression of the outcome named in the column heading, with the standard error in parentheses. The full control set is the child's sex, age in months and its square, birth order and twin status; the mother's age, parity and level of education; the father's education; the household's wealth quintile, urban residence, size and number of children under five; the age and sex of the household head; and the length of the household interview module. Every regression includes DHS cluster and day-of-week fixed effects, uses the DHS child weights, and clusters standard errors at the DHS cluster. Wasted, underweight and stunted are linear probability models for the indicator that the z-score falls below −2, with the coefficient scaled to percentage points. Values in square brackets are Anderson (2008) sharpened false-discovery-rate q-values, computed within the family of the sixteen block coefficients. *** p < 0.01, ** p < 0.05, * p < 0.1.*

Table 16 : Alternative definition of the treatment: equal-size blocks of the pooled start-time distribution (median, tertiles, quintiles)

	Weight-based outcomes					Height-based outcomes		
Block / statistic	**(1) Wasted (pp)**	**(2) WHZ**	**(3) Underweight (pp)**	**(4) WAZ**	**(5) Weight (kg)**	**(6) Stunted (pp)**	**(7) HAZ**	**(8) Height (cm)**
Below / above the median: 06:00–11:27 / 11:27–22:58 (children per block 56,970 / 57,082)								
Block 2 (11:27–22:58) vs block 1	−0.158 (0.193) [1.000]	0.020* (0.010) [0.762]	−0.249 (0.314) [1.000]	0.013 (0.010) [1.000]	0.038** (0.015) [0.179]	0.097 (0.394) [1.000]	0.001 (0.012) [1.000]	0.026 (0.041) [1.000]
Joint Wald p, all blocks = 0	0.413	0.059	0.428	0.203	0.010	0.807	0.924	0.526
Observations	114,052	114,052	114,489	114,489	114,550	114,550	114,550	114,550
Tertiles: 06:00–10:24 / 10:24–12:44 / 12:44–22:58 (children per block 38,012 / 37,910 / 38,130)								
Block 2 (10:24–12:44) vs block 1	−0.110 (0.229) [1.000]	0.014 (0.012) [1.000]	−0.036 (0.368) [1.000]	0.010 (0.012) [1.000]	0.018 (0.017) [1.000]	−0.354 (0.485) [1.000]	0.001 (0.014) [1.000]	0.002 (0.049) [1.000]
Block 3 (12:44–22:58) vs block 1	−0.459* (0.243) [0.762]	0.032** (0.013) [0.179]	−0.350 (0.387) [1.000]	0.020 (0.013) [1.000]	0.047*** (0.018) [0.179]	0.177 (0.493) [1.000]	−0.001 (0.015) [1.000]	0.018 (0.051) [1.000]
Joint Wald p, all blocks = 0	0.138	0.046	0.601	0.274	0.034	0.508	0.980	0.927
Observations	114,052	114,052	114,489	114,489	114,550	114,550	114,550	114,550
Quintiles: 06:00–09:34 / 09:34–10:49 / 10:49–12:10 / 12:10–14:12 / 14:12–22:58 (children per block 22,792 / 22,799 / 22,540 / 22,995 / 22,926)								
Block 2 (09:34–10:49) vs block 1	0.110 (0.300) [1.000]	0.000 (0.016) [1.000]	0.281 (0.477) [1.000]	−0.013 (0.015) [1.000]	−0.007 (0.022) [1.000]	−0.690 (0.622) [1.000]	−0.019 (0.018) [1.000]	−0.054 (0.064) [1.000]
Block 3 (10:49–12:10) vs block 1	−0.009 (0.298) [1.000]	0.021 (0.016) [1.000]	0.536 (0.478) [1.000]	−0.001 (0.016) [1.000]	0.010 (0.022) [1.000]	−0.275 (0.653) [1.000]	−0.024 (0.019) [1.000]	−0.075 (0.067) [1.000]
Block 4 (12:10–14:12) vs block 1	−0.115 (0.300)	0.012 (0.016)	0.322 (0.499)	0.001 (0.016)	0.017 (0.023)	−0.259 (0.634)	−0.009 (0.019)	−0.012 (0.067)

	Weight-based outcomes					Height-based outcomes		
Block / statistic	**(1) Wasted (pp)**	**(2) WHZ**	**(3) Underweight (pp)**	**(4) WAZ**	**(5) Weight (kg)**	**(6) Stunted (pp)**	**(7) HAZ**	**(8) Height (cm)**
	[1.000]	[1.000]	[1.000]	[1.000]	[1.000]	[1.000]	[1.000]	[1.000]
Block 5 (14:12–22:58) vs block 1	−0.495 (0.310) [1.000]	0.042** (0.017) [0.179]	−0.529 (0.499) [1.000]	0.022 (0.016) [1.000]	0.060** (0.024) [0.179]	0.068 (0.633) [1.000]	−0.008 (0.019) [1.000]	0.004 (0.067) [1.000]
Joint Wald p, all blocks = 0	0.346	0.073	0.226	0.283	0.061	0.766	0.724	0.669
Observations	114,052	114,052	114,489	114,489	114,550	114,550	114,550	114,550

***Notes**: All children aged 0–59 months, main specification (full control set, DHS cluster and day-of-week fixed effects, DHS child weights, standard errors clustered at the DHS cluster), with the treatment defined as the blocks named in the panel heading; block 1, the earliest, is omitted. Each block row is the coefficient relative to block 1 with the standard error in parentheses and, in square brackets, the Anderson (2008) sharpened q-value computed within the family of the panel's block coefficients across the eight outcomes; the joint Wald p-value tests that all block coefficients are zero. Wasted, underweight and stunted are linear probability models with coefficients in percentage points. Cut points are the quantiles of the pooled distribution of the household interview start time over all children in the analytical sample, so each block holds the same number of children. *** p < 0.01, ** p < 0.05, * p < 0.1.*

Table 17 : Alternative definition of the treatment: equal-duration blocks over the observed 06:00–22:58 range (halves, thirds, fifths)

	Weight-based outcomes					Height-based outcomes		
Block / statistic	**(1) Wasted (pp)**	**(2) WHZ**	**(3) Underweight (pp)**	**(4) WAZ**	**(5) Weight (kg)**	**(6) Stunted (pp)**	**(7) HAZ**	**(8) Height (cm)**
Halves: 06:00–14:29 / 14:29–22:58 (children per block 93,376 / 20,676)								
Block 2 (14:29–22:58) vs block 1	−0.466* (0.255) [0.191]	0.031** (0.014) [0.119]	−0.862** (0.413) [0.147]	0.027** (0.014) [0.151]	0.058*** (0.021) [0.032]	0.270 (0.513) [0.669]	0.011 (0.016) [0.540]	0.063 (0.056) [0.416]
Joint Wald p, all blocks = 0	0.067	0.025	0.037	0.044	0.005	0.599	0.468	0.257
Observations	114,052	114,052	114,489	114,489	114,550	114,550	114,550	114,550
Thirds: 06:00–11:39 / 11:39–17:19 / 17:19–22:58 (children per block 60,622 / 49,261 / 4,169)								
Block 2 (11:39–17:19) vs block 1	−0.169 (0.195) [0.469]	0.016 (0.011) [0.258]	−0.036 (0.319) [0.879]	0.010 (0.010) [0.444]	0.034** (0.015) [0.105]	0.229 (0.400) [0.669]	−0.002 (0.012) [0.879]	0.016 (0.041) [0.730]
Block 3 (17:19–22:58) vs block 1	−0.453 (0.551) [0.469]	0.042 (0.027) [0.257]	−1.641* (0.838) [0.155]	0.078*** (0.026) [0.020]	0.143*** (0.046) [0.020]	−1.874* (0.964) [0.155]	0.098*** (0.032) [0.020]	0.378*** (0.113) [0.019]
Joint Wald p, all blocks = 0	0.560	0.147	0.139	0.009	0.002	0.086	0.005	0.004
Observations	114,052	114,052	114,489	114,489	114,550	114,550	114,550	114,550
Fifths: 06:00–09:24 / 09:24–12:47 / 12:47–16:11 / 16:11–19:34 / 19:34–22:58 (children per block 20,141 / 56,620 / 28,245 / 8,580 / 466)								
Block 2 (09:24–12:47) vs block 1	0.039 (0.271) [0.879]	0.019 (0.014) [0.348]	0.027 (0.431) [0.879]	−0.000 (0.014) [0.897]	0.001 (0.020) [0.879]	−0.556 (0.571) [0.465]	−0.019 (0.017) [0.416]	−0.087 (0.060) [0.274]
Block 3 (12:47–16:11) vs block 1	−0.282 (0.307) [0.465]	0.028* (0.016) [0.207]	0.130 (0.493) [0.836]	−0.002 (0.016) [0.879]	0.011 (0.023) [0.697]	0.220 (0.635) [0.748]	−0.033* (0.019) [0.207]	−0.107 (0.067) [0.252]
Block 4 (16:11–19:34) vs block 1	−0.968** (0.441) [0.125]	0.076*** (0.023) [0.019]	−1.794** (0.709) [0.065]	0.071*** (0.024) [0.021]	0.127*** (0.037) [0.019]	−1.720** (0.846) [0.151]	0.043 (0.028) [0.258]	0.145 (0.100) [0.274]
Block 5 (19:34–22:58) vs block 1	0.734 (1.773) [0.730]	0.069 (0.080) [0.469]	−0.188 (2.439) [0.879]	0.095 (0.078) [0.372]	0.167 (0.123) [0.304]	−0.308 (2.777) [0.879]	0.075 (0.091) [0.469]	0.261 (0.320) [0.469]
Joint Wald p, all blocks = 0	0.113	0.022	0.063	0.015	0.008	0.117	0.048	0.055

	Weight-based outcomes					Height-based outcomes		
Block / statistic	**(1) Wasted (pp)**	**(2) WHZ**	**(3) Underweight (pp)**	**(4) WAZ**	**(5) Weight (kg)**	**(6) Stunted (pp)**	**(7) HAZ**	**(8) Height (cm)**
Observations	114,052	114,052	114,489	114,489	114,550	114,550	114,550	114,550

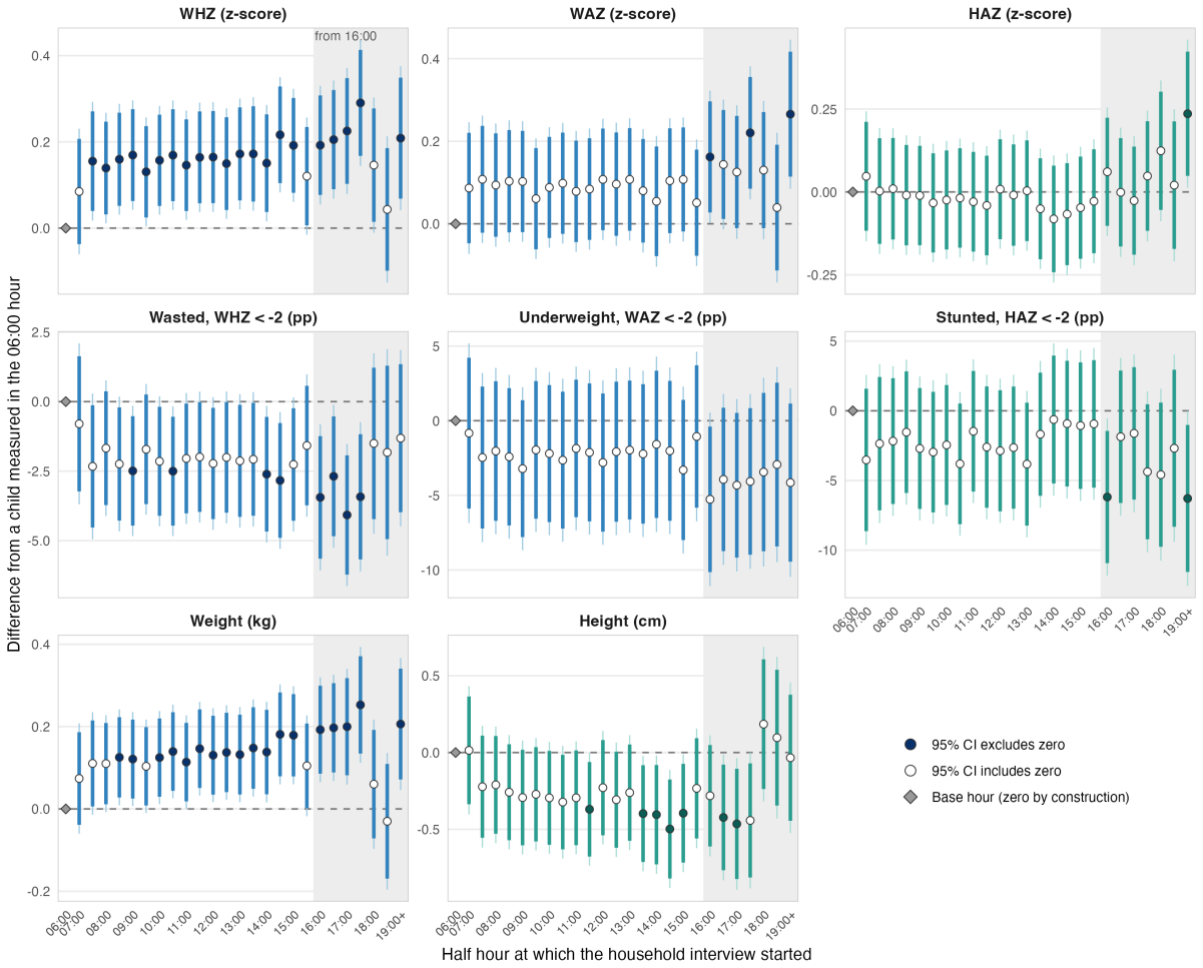
***Notes**: All children aged 0–59 months, main specification (full control set, DHS cluster and day-of-week fixed effects, DHS child weights, standard errors clustered at the DHS cluster), with the treatment defined as the blocks named in the panel heading; block 1, the earliest, is omitted. Each block row is the coefficient relative to block 1 with the standard error in parentheses and, in square brackets, the Anderson (2008) sharpened q-value computed within the family of the panel's block coefficients across the eight outcomes; the joint Wald p-value tests that all block coefficients are zero. Wasted, underweight and stunted are linear probability models with coefficients in percentage points. Cut points divide the observed range of start times, 06:00 to 22:58, into bands of equal length, so the late bands hold few children. *** p < 0.01, ** p < 0.05, * p < 0.1.*

Table 18 : Alternative definition of the treatment: the latest block relative to the earliest under ten partitions of the fieldwork day

		Weight-based outcomes					Height-based outcomes		
Partition	**Latest block**	**(1) Wasted (pp)**	**(2) WHZ**	**(3) Underweight (pp)**	**(4) WAZ**	**(5) Weight (kg)**	**(6) Stunted (pp)**	**(7) HAZ**	**(8) Height (cm)**
Main definition: 12:00 and 16:00 (3 blocks)	from 16:00, N = 10,110	−0.951** (0.373) [0.039]	0.052*** (0.020) [0.027]	−2.111*** (0.598) [0.001]	0.071*** (0.020) [0.002]	0.124*** (0.033) [0.001]	−1.474** (0.721) [0.031]	0.067*** (0.024) [0.011]	0.229*** (0.084) [0.023]
12:00 and 16:30 (3 blocks)	from 16:30, N = 7,539	−0.755* (0.410) [0.154]	0.056** (0.022) [0.032]	−1.662*** (0.644) [0.034]	0.067*** (0.022) [0.008]	0.114*** (0.036) [0.003]	−0.463 (0.756) [0.659]	0.058** (0.025) [0.057]	0.199** (0.090) [0.088]
12:00 and 17:00 (3 blocks)	from 17:00, N = 5,451	−0.809* (0.460) [0.169]	0.056** (0.026) [0.058]	−1.614** (0.742) [0.094]	0.070*** (0.025) [0.018]	0.126*** (0.042) [0.004]	−0.963 (0.861) [0.371]	0.074*** (0.029) [0.030]	0.266** (0.103) [0.036]
12:00 and 18:00 (3 blocks)	from 18:00, N = 2,293	0.692 (0.787) [0.153]	−0.020 (0.035) [0.074]	−1.046 (1.086) [0.476]	0.055* (0.033) [0.113]	0.133** (0.068) [0.013]	−2.031 (1.291) [0.216]	0.149*** (0.042) [0.002]	0.569*** (0.152) [0.001]
Below / above the median (2 blocks)	from 11:27, N = 57,348	−0.158 (0.193) [0.413]	0.020* (0.010) [0.059]	−0.249 (0.314) [0.428]	0.013 (0.010) [0.203]	0.038** (0.015) [0.010]	0.097 (0.394) [0.807]	0.001 (0.012) [0.924]	0.026 (0.041) [0.526]
Tertiles (3 blocks)	from 12:44, N = 38,315	−0.459* (0.243) [0.138]	0.032** (0.013) [0.046]	−0.350 (0.387) [0.601]	0.020 (0.013) [0.274]	0.047*** (0.018) [0.034]	0.177 (0.493) [0.508]	−0.001 (0.015) [0.980]	0.018 (0.051) [0.927]
Quintiles (5 blocks)	from 14:12, N = 23,037	−0.495 (0.310) [0.346]	0.042** (0.017) [0.073]	−0.529 (0.499) [0.226]	0.022 (0.016) [0.283]	0.060** (0.024) [0.061]	0.068 (0.633) [0.766]	−0.008 (0.019) [0.724]	0.004 (0.067) [0.669]
Halves (2 blocks)	from 14:29, N = 20,773	−0.466* (0.255) [0.067]	0.031** (0.014) [0.025]	−0.862** (0.413) [0.037]	0.027** (0.014) [0.044]	0.058*** (0.021) [0.005]	0.270 (0.513) [0.599]	0.011 (0.016) [0.468]	0.063 (0.056) [0.257]
Thirds (3 blocks)	from 17:19, N = 4,189	−0.453 (0.551) [0.560]	0.042 (0.027) [0.147]	−1.641* (0.838) [0.139]	0.078*** (0.026) [0.009]	0.143*** (0.046) [0.002]	−1.874* (0.964) [0.086]	0.098*** (0.032) [0.005]	0.378*** (0.113) [0.004]
Fifths (5 blocks)	from 19:34, N = 466	0.734 (1.773) [0.113]	0.069 (0.080) [0.022]	−0.188 (2.439) [0.063]	0.095 (0.078) [0.015]	0.167 (0.123) [0.008]	−0.308 (2.777) [0.117]	0.075 (0.091) [0.048]	0.261 (0.320) [0.055]

***Notes**: All children aged 0–59 months, main specification. Each row is one way of cutting the household interview start time into blocks: fixed clock cuts keep the morning block at 06:00–11:59 and move the start of the evening block; equal-size blocks are cut at the quantiles of the pooled start-time distribution; equal-duration blocks divide the observed 06:00–22:58 range into bands of equal length. The cell is the coefficient of the latest block relative to the earliest, with the standard error in parentheses and, in square brackets, the p-value of the joint Wald test that all block coefficients are zero. "Latest block" gives the clock time at which it starts and the number of children it holds. The full block-by-block estimates are in the three preceding tables. *** p < 0.01, ** p < 0.05, * p < 0.1.*

Figure 6 : Recorded anthropometry by the half hour at which the household interview started, all children aged 0–59 months



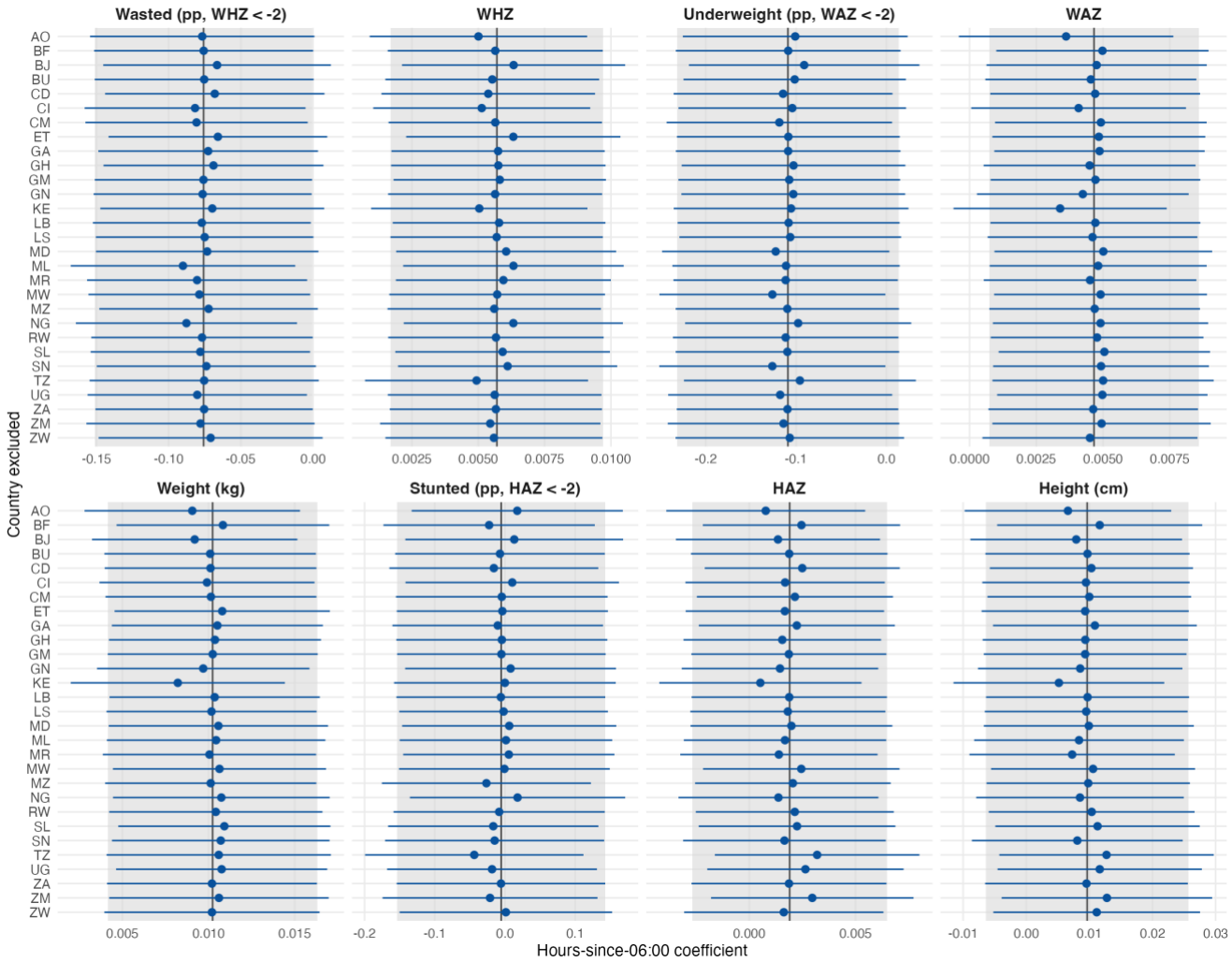
Notes: Estimates among all children aged 0–59 months and the full-control specification. One indicator for each half hour in which the household interview started, with the 06:00 hour omitted as the base; half hours with fewer than 500 interviews are pooled into the last point, 19:00 and later. Thick bars are ninety percent and thin bars ninety-five percent confidence intervals from standard errors clustered at the DHS cluster; filled points are estimates whose ninety-five percent interval excludes zero. The control set is that of Table 4; the weight equation also controls for height and the height equation for weight. DHS cluster and day-of-week fixed effects; survey weights. The shaded band marks interviews started at 16:00 or later.

Table 19 : Robustness to the clustering of the standard errors.

	Weight-based outcomes					Height-based outcomes		
Variance estimator	**(1) Wasted (pp)**	**(2) WHZ**	**(3) Underweight (pp)**	**(4) WAZ**	**(5) Weight (kg)**	**(6) Stunted (pp)**	**(7) HAZ**	**(8) Height (cm)**
Panel A. Clustered on the exact interview start minute								
Hours since 06:00	−0.076** (0.038) [0.063]	0.006*** (0.002) [0.015]	−0.109* (0.063) [0.093]	0.005** (0.002) [0.043]	0.010*** (0.003) [0.008]	−0.005 (0.076) [0.483]	0.002 (0.002) [0.215]	0.010 (0.008) [0.156]
Mean of outcome	4.961	−0.163	13.289	−0.826	11.168	27.754	−1.251	83.216
R^2	0.203	0.293	0.243	0.332	0.857	0.293	0.350	0.927
Observations	114,052	114,052	114,489	114,489	114,550	114,550	114,550	114,550
Panel B. Two-way clustering on the DHS cluster and the start minute								
Hours since 06:00	−0.076* (0.039) [0.074]	0.006*** (0.002) [0.021]	−0.109* (0.066) [0.109]	0.005** (0.002) [0.054]	0.010*** (0.003) [0.011]	−0.005 (0.077) [0.505]	0.002 (0.002) [0.221]	0.010 (0.008) [0.186]
Mean of outcome	4.961	−0.163	13.289	−0.826	11.168	27.754	−1.251	83.216
R^2	0.203	0.293	0.243	0.332	0.857	0.293	0.350	0.927
Observations	114,052	114,052	114,489	114,489	114,550	114,550	114,550	114,550

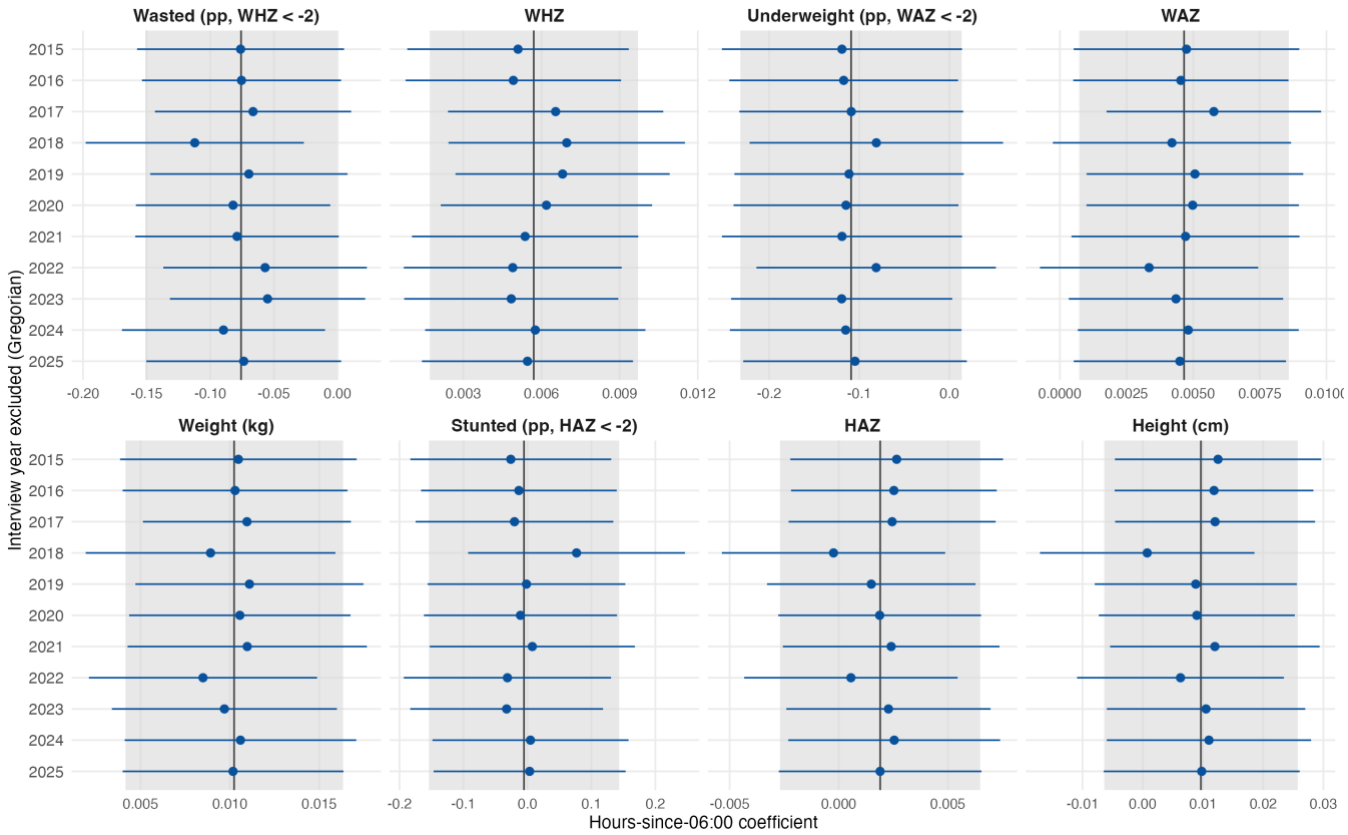
***Notes**: All children aged 0–59 months. Point estimates are those of the main specification; only the variance estimator changes. The full control set is the child's sex, age in months and its square, birth order and twin status; the mother's age, parity and level of education; the father's education; the household's wealth quintile, urban residence, size and number of children under five; the age and sex of the household head; and the length of the household interview module. Every regression includes DHS cluster and day-of-week fixed effects, uses the DHS child weights, and clusters standard errors at the DHS cluster. Wasted, underweight and stunted are linear probability models for the indicator that the z-score falls below −2, with the coefficient scaled to percentage points. Values in square brackets are Anderson (2008) sharpened false-discovery-rate q-values, computed within the family of the eight outcomes of the panel. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.*

Figure 7 : Leave-one-country-out estimates of the timing gradient.



***Notes**: All children aged 0–59 months, full-control specification. Each row is one of the 29 excluded countries, ordered alphabetically by ISO code. The point is the coefficient with that country dropped and the horizontal line its ninety-five percent confidence interval; the shaded band is the confidence interval of the full-sample estimate and the vertical line the full-sample point estimate. DHS cluster and day-of-week fixed effects, survey weights, standard errors clustered at the DHS cluster.*

Figure 8 : Leave-one-year-out estimates of the timing gradient.



***Notes**: All children aged 0–59 months, full-control specification. Each row is one excluded Gregorian interview year, earliest at the top. The point is the coefficient with that year dropped and the horizontal line its ninety-five percent confidence interval; the shaded band is the confidence interval of the full-sample estimate and the vertical line the full-sample point estimate. Ethiopian-calendar dates are converted before the split, so the year is comparable across surveys. DHS cluster and day-of-week fixed effects, survey weights, standard errors clustered at the DHS cluster.*

Table 20 : Covariate balance before and after entropy weighting.

Covariate	Standardized difference, before	Standardized difference, after
Female	−0.003	0.000
Age (months)	0.014	0.000
Birth order	−0.020	−0.000
Mother's age	0.011	−0.000
Parity	−0.021	−0.000
Mother's education: primary	−0.020	−0.000
Mother's education: secondary	−0.002	−0.000
Mother's education: higher	0.045	0.000
Wealth quintile	0.031	−0.000
Urban	0.072	0.000
Household size	−0.044	−0.000
No. children < 5	−0.031	−0.000
Head's age	−0.012	−0.000
Female-headed household	0.016	0.000

***Notes**: All children aged 0–59 months. Standardized mean difference between children whose interview started in the afternoon or evening and those whose interview started in the morning, weighted by the survey weight (before) and by the entropy-balancing weight (after). Entropy balancing reweights the late group so that the first moments of every covariate match the morning group exactly.*

Table 21 : Regression adjustment against entropy balancing.

	Weight-based outcomes					Height-based outcomes		
Estimator	**(1) Wasted (pp)**	**(2) WHZ**	**(3) Underweight (pp)**	**(4) WAZ**	**(5) Weight (kg)**	**(6) Stunted (pp)**	**(7) HAZ**	**(8) Height (cm)**
Panel A. Regression adjustment (main specification)								
Hours since 06:00	−0.076** (0.038) [0.065]	0.006*** (0.002) [0.018]	−0.109* (0.063) [0.089]	0.005** (0.002) [0.042]	0.010*** (0.003) [0.008]	−0.005 (0.076) [0.467]	0.002 (0.002) [0.217]	0.010 (0.008) [0.150]
Mean of outcome	4.961	−0.163	13.289	−0.826	11.168	27.754	−1.251	83.216
R^2	0.203	0.293	0.243	0.332	0.857	0.293	0.350	0.927
Observations	114,052	114,052	114,489	114,489	114,550	114,550	114,550	114,550
Panel B. Entropy balancing (covariates reweighted)								
Hours since 06:00	−0.079** (0.039) [0.055]	0.006*** (0.002) [0.013]	−0.118* (0.063) [0.065]	0.005** (0.002) [0.028]	0.011*** (0.003) [0.007]	−0.011 (0.076) [0.380]	0.002 (0.002) [0.184]	0.010 (0.008) [0.115]
Mean of outcome	4.961	−0.163	13.289	−0.826	11.168	27.754	−1.251	83.216
R^2	0.205	0.296	0.244	0.335	0.857	0.295	0.353	0.927
Observations	114,052	114,052	114,489	114,489	114,550	114,550	114,550	114,550

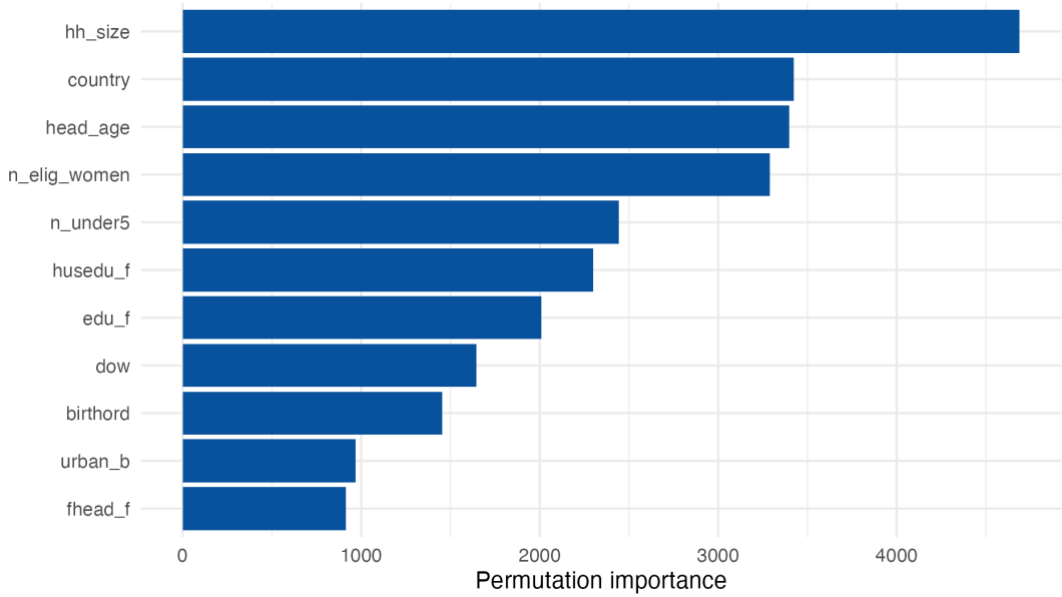
***Notes**: All children aged 0–59 months. Panel A is the main specification. Panel B re-estimates it with the entropy-balancing weights of Table 18 multiplied into the survey weights, so that the covariate distribution of children interviewed later in the day matches that of children interviewed in the morning. The full control set is the child's sex, age in months and its square, birth order and twin status; the mother's age, parity and level of education; the father's education; the household's wealth quintile, urban residence, size and number of children under five; the age and sex of the household head; and the length of the household interview module. Every regression includes DHS cluster and day-of-week fixed effects, uses the DHS child weights, and clusters standard errors at the DHS cluster. Wasted, underweight and stunted are linear probability models for the indicator that the z-score falls below −2, with the coefficient scaled to percentage points. Values in square brackets are Anderson (2008) sharpened false-discovery-rate q-values, computed within the family of the eight outcomes of the panel for each treatment definition. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.*

Table 22 : Quality of the random forest model for the imputed start time

Quantity	Value
Training records	229,222
Predictors	11 (interview structure + parental education)
10-fold cross-validated RMSE (minutes)	109.4
10-fold cross-validated MAE (minutes)	74.7
Hold-out RMSE (minutes)	109.1
Hold-out MAE (minutes)	74.5
Pre-2015 records with a valid women's-module start time	1,026,111
Predictable (structural features present and country/day-of-week seen in training)	684,573
After the analytic filters (measured, first visit, same day, valid outcomes, 06:00–21:00)	292,805
Observed timing (2015+, hv801 recovered): children (of whom aged 24–59 months; surveys)	114,996 (61,627; 40)
Added by imputation (pre-2015): children (of whom aged 24–59 months; surveys)	292,805 (162,033; 80)
Pooled total: children (of whom aged 24–59 months; surveys)	407,801 (223,660; 120)

***Notes**: Random forest (ranger) trained on the 2015+ surveys to predict the gap between the women's-module start time and the household start time, with an 80/20 hold-out split and 10-fold cross-validation within the training part. Predictors are the interview structure (household size, number of eligible women, number of children under five, position of the mother's line) and parental education. The imputed household start time for the pre-2015 surveys is the women's-module start time minus the predicted gap.*

Figure 9 : Permutation variable importance in the random forest gap model



***Notes**: Permutation variable importance from the gap model, measured as the increase in prediction error when each predictor is randomly permuted. Only interview-structure variables are used as predictors. The ordering shows which features of the fieldwork day carry the information the model relies on, and confirms that no anthropometric or nutrition variable contributes to the imputation.*

Table 23 : Extending the sample with machine-learning (random forest) imputed timing.

	Weight-based outcomes					Height-based outcomes		
Sample	**(1) Wasted (pp)**	**(2) WHZ**	**(3) Underweight (pp)**	**(4) WAZ**	**(5) Weight (kg)**	**(6) Stunted (pp)**	**(7) HAZ**	**(8) Height (cm)**
Panel A. Pooled observed and imputed timing, source of the timing controlled								
Hours since 06:00	−0.004 (0.022) [0.665]	0.004*** (0.001) [0.001]	−0.071** (0.031) [0.026]	0.003*** (0.001) [0.009]	0.006*** (0.001) [0.001]	0.012 (0.037) [0.665]	0.000 (0.001) [0.665]	0.005 (0.004) [0.250]
Mean of outcome	7.959	−0.163	18.110	−0.957	10.817	36.386	−1.472	81.572
R^2	0.186	0.235	0.208	0.286	0.834	0.274	0.322	0.895
Observations	402,611	402,611	403,048	403,048	403,109	406,942	406,942	406,942
Panel B. Placebo on the 2015+ surveys, where the true start time is observed								
True start time	−0.030 (p = 0.415)	0.003 (p = 0.081)	−0.109 (p = 0.041)	0.004 (p = 0.049)	0.006 (p = 0.034)	−0.007 (p = 0.912)	0.002 (p = 0.302)	0.010 (p = 0.214)
Out-of-fold imputed start time	−0.037 (p = 0.286)	0.004 (p = 0.017)	−0.124 (p = 0.015)	0.006 (p = 0.001)	0.010 (p = 0.000)	−0.042 (p = 0.510)	0.003 (p = 0.129)	0.014 (p = 0.050)
Ratio, imputed to true	1.24	1.29	1.14	1.58	1.65	5.72	1.36	1.46

***Notes**: All children aged 0–59 months. Panel A pools the 40 surveys with an observed household start time and the 80 pre-2015 surveys for which the start time is imputed by a random forest from the women's-module start time and the household's interview structure (Table 21), with an indicator for the source of the timing. Panel B estimates the hours-since-06:00 gradient on the same 2015+ children twice, once with the observed start time and once with the out-of-fold imputed one; the ratio is the attenuation (or inflation) that the imputation error produces. The full control set is the child's sex, age in months and its square, birth order and twin status; the mother's age, parity and level of education; the father's education; the household's wealth quintile, urban residence, size and number of children under five; the age and sex of the household head; and the length of the household interview module. Every regression includes DHS cluster and day-of-week fixed effects, uses the DHS child weights, and clusters standard errors at the DHS cluster. Wasted, underweight and stunted are linear probability models for the indicator that the z-score falls below −2, with the coefficient scaled to percentage points. Values in square brackets are Anderson (2008) sharpened false-discovery-rate q-values, computed within the family of the eight outcomes of the panel for each treatment definition. *** p < 0.01, ** p < 0.05, * p < 0.1.*

Table 24 : The MICS replication sample.

Country	Years	Children	Clusters	Aged 24–59 months	Aged 0–23 months	Mean start hour	Also in DHS sample
Chad	2019	13,914	765	9,137	4,777	11:29	NO -- new country
Malawi	2019-2020	11,259	1,110	6,578	4,681	11:33	yes
Madagascar	2018	9,509	769	4,578	4,931	11:21	yes
Gambia	2018	7,110	387	4,397	2,713	13:18	yes
Sierra Leone	2017	6,766	568	4,126	2,640	12:19	yes
Ghana	2017	5,523	643	3,363	2,160	11:07	yes
Central Africa Republic	2018-2019	4,646	434	2,854	1,792	10:49	NO -- new country
Zimbabwe	2019	4,011	457	2,295	1,716	11:27	yes
Togo	2017	3,536	407	2,100	1,436	11:26	NO -- new country
Lesotho	2018	1,981	388	1,076	905	12:01	yes
Eswatini	2021-2022	1,569	313	910	659	11:46	NO -- new country

***Notes**: The eleven MICS surveys used in the replication, ordered by the number of children contributing valid anthropometry and a usable household interview start time. "Standing" and "recumbent" give the split at 24 months of age that the DHS protocol also applies. Rounds whose fieldwork crossed a calendar year are shown with both years. "Mean start hour" is the unweighted mean clock time at which the household interview began. The final column records whether the country also contributes a survey to the DHS analytic sample; where it does, the MICS round is a different survey in a different year with a different questionnaire and different field staff.*

Table 25 : Replication in the Multiple Indicator Cluster Surveys.

	Weight-based outcomes					Height-based outcomes		
Treatment / sample	**(1) Wasted (pp)**	**(2) WHZ**	**(3) Underweight (pp)**	**(4) WAZ**	**(5) Weight (kg)**	**(6) Stunted (pp)**	**(7) HAZ**	**(8) Height (cm)**
All children								
Hours since 06:00	−0.062 (0.045) [0.157]	0.005** (0.002) [0.093]	−0.148** (0.073) [0.093]	0.005** (0.002) [0.093]	0.007** (0.003) [0.093]	−0.031 (0.083) [0.462]	0.001 (0.003) [0.462]	0.002 (0.009) [0.462]
Mean of outcome	7.135	−0.315	18.555	−1.008	11.179	31.363	−1.370	83.940
R^2	0.133	0.212	0.159	0.229	0.807	0.192	0.230	0.894
Observations	69,309	69,309	69,721	69,721	69,824	69,824	69,824	69,824

***Notes**: MICS surveys of Table 24, estimated with the MICS household interview start time as the treatment, the MICS analogues of the full control set, MICS cluster and day-of-week fixed effects, survey weights and standard errors clustered at the MICS cluster. Wasted, underweight and stunted are linear probability models for the indicator that the z-score falls below −2, with the coefficient scaled to percentage points. Values in square brackets are Anderson (2008) sharpened false-discovery-rate q-values, computed within the family of the eight outcomes of the panel for each treatment definition. *** p < 0.01, ** p < 0.05, * p < 0.1.*

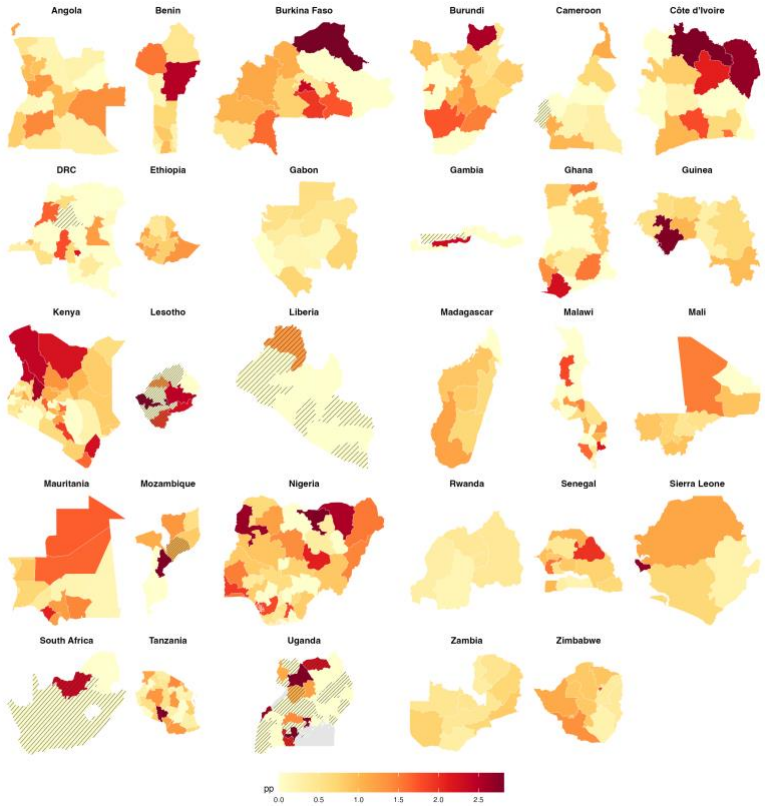
Table 26 : Children under five mis-classified as adequately nourished after anchoring measurement to 06:00, by country

				Underweight (WAZ < −2)		Wasted (WHZ < −2)		Either	
Country	**Rounds**	**Children 0–5 (N)**	**Population under five**	**Mis-classified, % [95% CI]**	**Missed children**	**Mis-classified, % [95% CI]**	**Missed children**	**Mis-classified, % [95% CI]**	**Missed children**
Angola	2	11,351	5,127,623	0.884 [0.344 – 1.317]	45,306	0.294 [0.128 – 0.374]	15,083	0.912 [0.380 – 1.328]	46,757
Benin	1	11,663	2,022,832	0.922 [0.222 – 1.709]	18,641	0.495 [0.148 – 0.760]	10,018	1.009 [0.265 – 1.817]	20,416
Burkina Faso	1	5,723	3,384,157	0.955 [0.334 – 1.381]	32,332	0.776 [0.290 – 1.075]	26,252	1.192 [0.486 – 1.645]	40,327
Burundi	1	6,047	2,141,808	1.331 [0.425 – 2.329]	28,507	0.312 [0.073 – 0.603]	6,691	1.368 [0.435 – 2.394]	29,299
Cameroon	1	4,492	4,044,315	0.405 [0.261 – 0.646]	16,390	0.349 [0.121 – 0.512]	14,126	0.619 [0.364 – 0.918]	25,016
Congo, Dem. Rep.	1	10,740	—	0.824 [0.201 – 1.320]	—	0.363 [0.127 – 0.481]	—	0.823 [0.252 – 1.327]	—
Côte d'Ivoire	1	4,774	4,549,135	1.035 [0.195 – 1.570]	47,062	0.745 [0.214 – 1.175]	33,870	1.467 [0.315 – 2.202]	66,745
Ethiopia	3	19,760	—	1.003 [0.273 – 1.617]	—	0.418 [0.227 – 0.697]	—	1.133 [0.397 – 1.812]	—
Gabon	1	5,340	315,740	0.170 [0.054 – 0.422]	536	0.167 [0.119 – 0.275]	529	0.316 [0.168 – 0.623]	999
Gambia	1	3,811	375,851	0.538 [0.094 – 1.346]	2,022	0.489 [0.279 – 0.899]	1,836	0.767 [0.237 – 1.505]	2,881
Ghana	1	4,404	4,204,477	0.605 [0.137 – 0.964]	25,419	0.197 [0.082 – 0.533]	8,289	0.551 [0.156 – 1.046]	23,173
Guinea	1	3,455	2,040,353	0.896 [0.278 – 1.339]	18,287	0.757 [0.343 – 1.102]	15,448	1.332 [0.563 – 1.891]	27,176
Kenya	1	17,323	6,981,963	0.596 [0.181 – 0.943]	41,602	0.291 [0.115 – 0.611]	20,290	0.656 [0.236 – 1.109]	45,827
Lesotho	1	1,093	266,338	0.593 [0.112 – 0.750]	1,580	0.236 [0.236 – 0.236]	629	0.594 [0.112 – 0.752]	1,583
Liberia	1	2,444	746,562	0.977 [0.271 – 1.406]	7,294	0.233 [0.174 – 0.262]	1,738	0.992 [0.272 – 1.277]	7,405
Madagascar	1	5,777	4,420,158	0.832 [0.193 – 1.501]	36,783	0.419 [0.143 – 0.701]	18,501	0.864 [0.223 – 1.584]	38,202
Malawi	2	10,554	2,870,172	0.557 [0.210 – 0.945]	15,991	0.107 [0.031 – 0.229]	3,067	0.602 [0.214 – 1.057]	17,290
Mali	2	22,229	3,800,689	0.820 [0.208 – 1.512]	31,156	0.502 [0.191 – 0.792]	19,088	0.960 [0.276 – 1.685]	36,479

				Underweight (WAZ < −2)		Wasted (WHZ < −2)		Either	
Country	**Rounds**	**Children 0–5 (N)**	**Population under five**	**Mis-classified, % [95% CI]**	**Missed children**	**Mis-classified, % [95% CI]**	**Missed children**	**Mis-classified, % [95% CI]**	**Missed children**
Mauritania	1	9,826	770,514	0.831 [0.252 – 1.501]	6,402	0.813 [0.269 – 1.230]	6,261	1.160 [0.359 – 1.887]	8,940
Mozambique	1	3,733	5,547,107	0.888 [0.109 – 1.569]	49,254	0.195 [0.012 – 0.477]	10,815	0.994 [0.110 – 1.811]	55,121
Nigeria	2	20,760	33,420,838	0.946 [0.236 – 1.687]	316,039	0.428 [0.185 – 0.688]	142,881	1.063 [0.301 – 1.872]	355,168
Rwanda	2	10,521	1,812,231	0.364 [0.159 – 0.591]	6,591	0.040 [0.011 – 0.099]	721	0.368 [0.164 – 0.603]	6,673
Senegal	4	26,801	2,451,638	0.910 [0.236 – 1.532]	22,304	0.743 [0.243 – 1.175]	18,216	1.161 [0.317 – 1.881]	28,466
Sierra Leone	1	4,133	1,115,078	1.026 [0.222 – 1.531]	11,441	0.515 [0.238 – 0.669]	5,738	1.200 [0.290 – 1.715]	13,378
South Africa	1	1,096	5,730,108	0.293 [0.000 – 0.360]	16,789	0.052 [0.000 – 0.052]	3,005	0.283 [0.000 – 0.351]	16,229
Tanzania	2	13,798	10,595,923	0.609 [0.136 – 1.035]	64,566	0.281 [0.089 – 0.436]	29,780	0.659 [0.174 – 1.084]	69,776
Uganda	1	4,415	6,839,230	0.513 [0.115 – 0.826]	35,098	0.169 [0.085 – 0.336]	11,531	0.561 [0.157 – 0.968]	38,377
Zambia	2	12,556	2,919,797	0.648 [0.190 – 1.070]	18,922	0.211 [0.062 – 0.275]	6,159	0.718 [0.199 – 1.153]	20,973
Zimbabwe	1	4,951	2,378,323	0.649 [0.213 – 0.913]	15,435	0.234 [0.147 – 0.466]	5,574	0.653 [0.182 – 1.010]	15,535
Total (27 countries with a population denominator)			**120,872,960**	**0.771**	**931,747**	**0.361**	**436,137**	**0.875**	**1,058,210**

***Notes**: DHS data only, all measured children aged 0–59 months with a valid z-score in the country's phase-7+ surveys (no first-visit, same-day or window restriction), rounds pooled with the round's sample size as weight. "Mis-classified" is the DHS-weighted share of children under five whose classification flips when their weight-based z-score is moved to its 06:00 value with the gradients of Table 4 (WAZ 0.005 and WHZ 0.006 per hour), i.e. the counterfactual prevalence at 06:00 minus the observed prevalence; the 95% confidence interval moves the gradient by 1.96 standard errors. "Either" counts a child once if it flips on underweight or on wasting. "Population under five" is the country's under-5 population (latest DHS round, reference years 2015–2022); "Missed children" is the mis-classified share applied to that population. Ethiopia and the Democratic Republic of Congo have no under-5 population figure in the panel and are shown without a count; the total covers the 27 countries with a denominator.*

Figure 10 : Percentage-point understatement of weight-based undernutrition prevalence among children aged 0–59 months, by admin-1 unit, country by country



***Notes**: Percentage-point understatement of the prevalence of undernutrition at the −2 threshold among children aged 0–59 months, by GADM 4.1 admin-1 unit (regions or provinces), with one panel per country drawn at its own scale. Each DHS cluster is assigned to a unit by its displaced GPS coordinates (2 km in urban and 5–10 km in rural areas), the national gradient of Table 4 is applied within the unit, and the bias is the survey-weighted difference between the observed prevalence and the prevalence after re-anchoring to 06:00, so that differences between units come from the local distribution of interview start hours and the share of children near the cut-off. Of the 477 admin-1 units with at least one cluster, 366 reach the minimum of 50 children aged 0–59 months and are coloured; hatched units fall below that minimum and light grey units have no DHS cluster. Lesotho has no admin-2 layer in GADM and is shown at admin-1. Counts are not mapped at this level because no sub-national population denominator exists.*